

CONTINUITY & DIFFERENTIABILITY

JEE EXERCISES

Exercise J-1

CONTINUITY CHECK

1. f is a continuous function on the real line. Given that

$x^2 + (f(x) - 2)x - \sqrt{3} \cdot f(x) + 2\sqrt{3} - 3 = 0$. Then the value of $f(\sqrt{3})$:

- (A) can not be determined (B) $2(1 - \sqrt{3})$
(C) zero (D) $\frac{2(\sqrt{3} - 2)}{\sqrt{3}}$

यदि प्रत्येक वास्तविक संख्या के लिये f एक सतत फलन हो तथा दिया है कि

$x^2 + (f(x) - 2)x - \sqrt{3} \cdot f(x) + 2\sqrt{3} - 3 = 0$ तो $f(\sqrt{3})$ का मान होगा :

- (A) ज्ञात नहीं किया जा सकता (B) $2(1 - \sqrt{3})$
(C) शून्य (D) $\frac{2(\sqrt{3} - 2)}{\sqrt{3}}$

Ans. B

Sol. $(x - \sqrt{3}) f(x) = -x^2 + 2x - 2\sqrt{3} + 3$

$$f(x) = \frac{-x^2 + 2x - 2\sqrt{3} + 3}{x - \sqrt{3}}$$
$$= \frac{(x - \sqrt{3})(2 - \sqrt{3} - x)}{x - \sqrt{3}} = 2 - \sqrt{3} - x$$

2. The function $f(x) = [x]^2 - [x^2]$ (where $[y]$ is the greatest integer less than or equal to y), is discontinuous at :

- (A) all integers (B) all integers except 0 & 1
(C) all integers except 0 (D) all integers except 1

फलन $f(x) = [x]^2 - [x^2]$ कहाँ – कहाँ असतत होगा (जहाँ $[y]$ महत्तम पूर्णांक फलन है y) :

- (A) सभी पूर्णांको पर (B) 0 व 1 को छोड़कर सभी पूर्णांको पर
(C) 0 के छोड़कर सभी पूर्णांको पर (D) 1 को छोड़कर सभी पूर्णांको पर

Ans. D

Sol. $f(x) = [x]^2 - [x^2]$

$$(-1)^2 - 0 = 1$$

$$x = 0 - h$$

$$f(x) = \begin{cases} 0 & x = 0 \\ 0 - 0 = 0 & 0 < x < 1 \\ 1^2 - 1 = 0 & 1 \leq x < \sqrt{2} \\ 1 - 2 = -1 & \sqrt{2} \leq x < \sqrt{3} \\ 1 - 3 = -2 & \sqrt{3} \leq x < 2 \end{cases}$$

$$\begin{aligned} [x] &= -1 \\ [x^2] &= 0 \\ 1 \leq x^2 < 2 \\ [x] &= 1 \\ 2 \leq x^2 < 3 \end{aligned}$$

i.e. it is continuous at $x = 1$ only and discontinuous at all other points]

3. Let $g(x) = \tan^{-1}|x| - \cot^{-1}|x|$, $f(x) = \frac{[x]}{[x+1]} \{x\}$, $h(x) = |g(f(x))|$ where $\{x\}$ denotes fractional part and $[x]$

denotes the integral part then which of the following holds good :

- (A) h is continuous at $x = 0$ (B) h is discontinuous at $x = 0$
 (C) $h(0^-) = \pi/2$ (D) $h(0^+) = -\pi/2$

माना कि $g(x) = \tan^{-1}|x| - \cot^{-1}|x|$, $f(x) = \frac{[x]}{[x+1]} \{x\}$, $h(x) = |g(f(x))|$ जहाँ $\{x\}$ एक अपूर्णाश फलन है तथा $[x]$ एक महत्तम पूर्णांक फलन हो, तो निम्न में से कौन सा विकल्प सत्य है :

- (A) $x = 0$ पर h सतत् है। (B) $x = 0$ पर h असतत् फलन है।
 (C) $h(0^-) = \pi/2$ (D) $h(0^+) = -\pi/2$

Ans. A

Sol. $g(x) = \tan^{-1}|x| - \cot^{-1}|x|$

$$f(x) = \frac{[x]}{[x+1]} \{x\}; x \in [-1, 0)$$

$$g(x) = |g(f(x))|$$

$$h(0^+) = \lim_{\alpha \rightarrow 0} h(0 + \alpha) = \lim_{\alpha \rightarrow 0} |g(f(\alpha))|$$

$$|g(0)| = \left| 0 - \frac{\pi}{2} \right| = \frac{\pi}{2}$$

$$h(0) = |g(f(0))| = |g(0)| = \frac{\pi}{2}$$

$\Rightarrow h$ is 0 at $x = 0$

4. The function $f(x) = [x] \cdot \cos \frac{2x-1}{2}\pi$, where $[\cdot]$ denotes the greatest integer function, is discontinuous at :

- (A) all x (B) all integer points (C) no x (D) x which is not an integer

दिया है कि $f(x) = [x] \cdot \cos \frac{2x-1}{2}\pi$, जहाँ $[\cdot]$ एक महत्तम पूर्णांक फलन हो, तो फलन $f(x)$ असतत् होगा :

- (A) सभी x पर (B) सभी पूर्णांक संख्या पर
 (C) किसी भी x पर नहीं (D) सभी x पर जो पूर्णांक नहीं हो

Ans. C

Sol. $f(x) = [x] \cos \left(\left(\frac{2x-1}{2} \right) \pi \right)$

$$\Rightarrow f(x) = [x] \cdot \cos \left(\pi x - \frac{\pi}{2} \right)$$

$$f(x) = [x] \sin \pi x$$

Let $x = k \in \mathbb{Z}$

$$\lim_{x \rightarrow k} f(x)$$

$$\text{L.H.L.} = \lim_{h \rightarrow 0} f(k-h) = \lim_{h \rightarrow 0} [k-h] \sin(\pi k - \pi h) = 0$$

$$\text{RHL} = \lim_{h \rightarrow 0} f(k+h) = \lim_{h \rightarrow 0} [k+h] \sin(\pi k + \pi h) = 0$$

$$f(k) = [k] \sin \pi k = 0$$

$\therefore f(x)$ is continuous $\forall x \in \mathbb{R}$

5. Consider the functions

$$f(x) = \text{sgn}(x-1) \text{ and } g(x) = \cot^{-1}[x-1]$$

where $[]$ denotes the greatest integer function :

Statement-1 : The function $F(x) = f(x) \cdot g(x)$ is discontinuous at $x = 1$.

because

Statement-2 : If $f(x)$ is discontinuous at $x = a$ and $g(x)$ is also discontinuous at $x = a$ then the product function $f(x) \cdot g(x)$ is discontinuous at $x = a$.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true

दिया है

$$f(x) = \text{sgn}(x-1) \text{ और } g(x) = \cot^{-1}[x-1]$$

जहाँ $[]$ महत्तम पूर्णांक फलन है :

कथन-1 : फलन $F(x) = f(x) \cdot g(x)$, $x = 1$ असतत् होगा

क्योंकि

कथन-2 : यदि $x = a$ पर फलन $f(x)$ असतत् है तथा $g(x)$ भी $x = a$ पर असतत् है तो उनका गुणनफल

$$f(x) \cdot g(x) \text{ भी } x = a \text{ पर असतत् होगा}$$

(A) कथन-1 सत्य है, कथन-2 सत्य है, कथन-2 कथन-1 का सही स्पष्टीकरण है।

(B) कथन-1 सत्य है, कथन-2 सत्य है, कथन-2 कथन-1 का सही स्पष्टीकरण नहीं है।

(C) कथन-1 सत्य है, कथन-2 असत्य है।

(D) कथन-1 असत्य है, कथन-2 सत्य है।

Ans. C

Sol. $f(x) = \operatorname{sgn}(x-1)$ $g(x) = \cot^{-1}[x-1]$

$$F(x) = f(x) \cdot g(x)$$

$$= \operatorname{sgn}(x-1) \cdot \cot^{-1}[x-1]$$

$$F(1) = 0$$

$$\text{L.H.L.} = \lim_{h \rightarrow 0} F(1-h) = \lim_{h \rightarrow 0} \operatorname{sgn}(-h) \cot^{-1}[-h]$$

$$= -\pi$$

$$\text{R.H.L.} = \lim_{h \rightarrow 0} F(1+h) = \lim_{h \rightarrow 0} \operatorname{sgn}(h) \cot^{-1}[h] = \frac{\pi}{2}$$

$\therefore F(x)$ is discontinuous at $x = 1$.

Statement - 2

$$F(x) = \begin{cases} \frac{1}{2} & 0 \leq x \leq 1 \\ 2 & 1 < x \leq 2 \end{cases} \quad g(x) = \begin{cases} 2 & 0 \leq x \leq 1 \\ \frac{1}{2} & 1 < x \leq 2 \end{cases} \rightarrow F(x) \text{ \& } g(x) \text{ are discontinuous at } x = 1 \text{ but } f(x) \cdot g(x) = 1 \forall x \in [0, 2]$$

$\Rightarrow F(x)$ is discontinuous at $x = 1$

TYPES OF DISCONTINUITY

$$6. \quad f(x) = \begin{cases} \sin\left(\frac{a-x}{2}\right) \tan\left[\frac{\pi x}{2a}\right] & \text{for } x > a \\ \frac{\cos\left(\frac{\pi x}{2a}\right)}{a-x} & \text{for } x < a \end{cases}$$

where $[x]$ is the greatest integer function of x , and $a > 0$, then :

(A) $f(a^-) < 0$

(B) f has a removable discontinuity at $x = a$

(C) f has an irremovable discontinuity at $x = a$

(D) $f(a^+) < 0$

$$f(x) = \begin{cases} \sin\left(\frac{a-x}{2}\right) \tan\left[\frac{\pi x}{2a}\right] & \text{for } x > a \\ \frac{\cos\left(\frac{\pi x}{2a}\right)}{a-x} & \text{for } x < a \end{cases}$$

जहाँ $[x]$ एक महत्तम पूर्णांक फलन हो, तथा $a > 0$ है तब :

- (A) $f(a^-) < 0$
 (B) $x = a$ पर f विस्थापनीय बिन्दु असत्ता प्रदर्शित करता है
 (C) $x = a$ पर f अविस्थापनीय बिन्दु असत्ता प्रदर्शित करता है
 (D) $f(a^+) < 0$

Ans. B

Sol.

$$f(x) = \begin{cases} \sin\left(\frac{a-x}{2}\right) \tan\left[\frac{\pi x}{2}\right] & \text{For } x > a \\ \frac{\cos\left(\frac{\pi x}{2a}\right)}{a-x} & \text{For } x < a \end{cases}$$

$$\lim_{h \rightarrow 0} f(a+h)$$

$$= \lim_{h \rightarrow 0} \sin\left(\frac{a-(a+h)}{2}\right) \tan\left[\frac{\pi a}{2a} + \frac{\pi h}{2a}\right]$$

$$\lim_{h \rightarrow 0} -\sin\left(\frac{h}{2}\right) \cdot \tan\left[\frac{\pi}{2} + \frac{h}{2a}\right] = 0$$

$$\lim_{h \rightarrow 0} f(a-h) = \lim_{h \rightarrow 0} \sin\left(\frac{h}{2}\right) \tan\left[\frac{\pi}{2} - \frac{\pi h}{2a}\right]$$

$$= 0$$

$\Rightarrow f$ has a removable discontinuity at $x = a$.

DIFFERENTIABILITY CHECK

7. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \text{Min} \{x+1, |x|+1\}$. Then which of the following is true :

- (A) $f(x) \geq 1$ for all $x \in \mathbb{R}$ (B) $f(x)$ is not differentiable at $x = 1$
 (C) $f(x)$ is differentiable everywhere (D) $f(x)$ is not differentiable at $x = 0$

माना $f: \mathbb{R} \rightarrow \mathbb{R}$ एक फलन है जो $f(x) = \text{Min} \{x+1, |x|+1\}$ से परिभाषित है, तब निम्न में से कौनसा सही है :

- (A) $f(x) \geq 1$ सभी $x \in \mathbb{R}$ के लिये (B) $f(x)$, $x = 1$ पर अवकलनीय नहीं है
 (C) $f(x)$, प्रत्येक बिन्दु पर अवकलनीय है (D) $f(x)$, $x = 0$ पर अवकलनीय नहीं है

Ans. C

Sol. $f(x) = \text{Min}[x+1, |x|+1]$

$$= \begin{cases} x+1 & \text{if } x \geq 0 \\ |x|+1 & \text{if } x < 0 \end{cases}$$

$$\Rightarrow f(x) = x+1$$

$\therefore f(x)$ is differentiable everywhere

8. Let $f(x) = \begin{cases} (x-1)\sin\frac{1}{x-1} & \text{if } x \neq 1 \\ 0 & \text{if } x = 1 \end{cases}$. Then which one of the following is true :

- (A) f is differentiable at $x=0$ and at $x=1$ (B) f is differentiable at $x=0$ but not at $x=1$
(C) f is differentiable at $x=1$ but not at $x=0$ (D) f is neither differentiable at $x=0$ nor at $x=1$

माना $f(x) = \begin{cases} (x-1)\sin\frac{1}{x-1} & \text{if } x \neq 1 \\ 0 & \text{if } x = 1 \end{cases}$ तो निम्न में से कौनसा सत्य है :

- (A) $f, x=0$ तथा $x=1$ पर अवकलनीय है (B) $f, x=0$ पर अवकलनीय है लेकिन $x=1$ पर नहीं
(C) $f, x=1$ पर अवकलनीय है लेकिन $x=0$ पर नहीं (D) f , न तो $x=0$ पर तथा न ही $x=1$ पर अवकलनीय है

Ans. B

Sol. $f(x) = \begin{cases} (x-1)\sin\frac{1}{(x-1)} & \text{if } x \neq 1 \\ 0 & x = 1 \end{cases}$

$$\lim_{x \rightarrow 1} f(x) = 0 = f(1)$$

$\therefore f(x)$ will be continuous at $x=1$.

$$\text{L.H.D.} = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h} = \lim_{h \rightarrow 0} \frac{h \sin \frac{1}{h}}{-h} = \text{D.N.E.}$$

$$\text{R.H.D.} = \lim_{h \rightarrow 0} \sin\left(\frac{1}{h}\right) = \text{DNE}$$

$\therefore f(x)$ is not differentiable at $x=1$.

but $f(x)$ is differentiable at $x=0$.

9. Let $g(x) = \frac{(x-1)^n}{\ln \cos^m(x-1)}$; $0 < x < 2$, m and n are integers, $m \neq 0$, $n > 0$ and let p be the left hand derivative of $|x-1|$ at $x=1$. If $\lim_{x \rightarrow 1^+} g(x) = p$, then :

- (A) $n=1, m=1$ (B) $n=1, m=-1$ (C) $n=2, m=2$ (D) $n>2, m=n$

यदि $g(x) = \frac{(x-1)^n}{\ln \cos^m(x-1)}$; $0 < x < 2$, m तथा n पूर्णांक है, $m \neq 0$, $n > 0$. माना कि $|x-1|$ का $x=1$ पर वाम पक्ष अवकलज

(LHD) p है, तो यदि $\lim_{x \rightarrow 1^+} g(x) = p$, तो :

- (A) $n = 1, m = 1$ (B) $n = 1, m = -1$ (C) $n = 2, m = 2$ (D) $n > 2, m = n$

Ans. C

Sol. $p = -1$ Now

$$\lim_{x \rightarrow 1^+} \frac{(x-1)^n}{\log \cos^m(x-1)} = \lim_{x \rightarrow 1^+} \frac{(x-1)^n}{\log[\cos^m(x-1) - 1 + 1]}$$

$$= \lim_{x \rightarrow 1^+} \frac{(x-1)^n}{\cos^m(x-1) - 1}$$

$$= \lim_{x \rightarrow 1^+} \frac{n(x-1)^{n-1}}{m \cos^{m-1}(x-1) \sin(x-1)}$$

$$= \frac{-n}{m} \lim_{x \rightarrow 1^+} \frac{(x-1)}{\sin(x-1)} \cdot \frac{1}{\cos^{m-1}(x-1)} \times (x-1)^{n-2}$$

$$= \frac{-n}{m} \lim_{x \rightarrow 1^+} (x-1)^{n-2} = -1 \text{ (Given)}$$

$$\Rightarrow n = 2 \text{ and } m = 2$$

10. For what triplets of real numbers (a, b, c) with $a \neq 0$ the function

$$f(x) = \begin{cases} x & x \leq 1 \\ ax^2 + bx + c & \text{otherwise} \end{cases} \text{ is differentiable for all real } x :$$

यदि फलन $f(x) = \begin{cases} x & x \leq 1 \\ ax^2 + bx + c & \text{otherwise} \end{cases}$ x के प्रत्येक मान के लिये अवकलनीय है। जहाँ $a \neq 0$ तो (a, b, c) का मान होगा :

- (A) $\{(a, 1-2a, a) \mid a \in \mathbb{R}, a \neq 0\}$ (B) $\{(a, 1-2a, c) \mid a, c \in \mathbb{R}, a \neq 0\}$
 (C) $\{(a, b, c) \mid a, b, c \in \mathbb{R}, a + b + c = 1\}$ (D) $\{(a, 1-2a, 0) \mid a \in \mathbb{R}, a \neq 0\}$

Ans. A

Sol. $f(x) = \begin{cases} x & , x \leq 1 \\ ax^2 + bx + c & , \text{otherwise} \end{cases}$

$f(x)$ should be continuous at $x=1$

it gives $a+b+c=1$

$f(x)$ should be differentiable at $x=1$

it gives $2a+b=1 \Rightarrow b=1-2a \quad c=1-a-b=a$

11. Number of points of non-differentiability of the function

$g(x) = [x^2] \{\cos^2 4x\} + \{x^2\} [\cos^2 4x] + x^2 \sin^2 4x + [x^2] [\cos^2 4x] + \{x^2\} \{\cos^2 4x\}$ in $(-50, 50)$ where $[x]$ and $\{x\}$ denote the greatest integer function and fractional part function of x respectively, is equal to :

फलन $g(x) = [x^2]\{\cos^2 4x\} + \{x^2\}[\cos^2 4x] + x^2 \sin^2 4x + [x^2][\cos^2 4x] + \{x^2\}\{\cos^2 4x\}$ के $(-50, 50)$ में उन बिन्दुओं की संख्या जहाँ $g(x)$ अवकलनीय नहीं है ($[x]$ तथा $\{x\}$ महत्तम पूर्णांक फलन तथा भिन्नांश भाग फलन का दर्शाता है) :

- (A) 98 (B) 99 (C) 100 (D) 0

Ans. D

Sol. $g(x) = ([x^2] + \{x^2\}) (\{\cos^2 4x\} + [\cos^2 4x]) + x^2 \sin^2 4x = x^2 \cos^2 4x + x^2 \sin^2 4x = x^2$

Clearly $g(x)$ is continuous and differentiable $\forall x \in \mathbb{R}$.]

12. If $f(x) = \begin{cases} x + \{x\} + x \sin \{x\} & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$ where $\{x\}$ denotes the fractional part function, then :

- (A) 'f' is continuous & diff. at $x = 0$
 (B) 'f' is continuous but not diff. at $x = 0$
 (C) 'f' is continuous & diff. at $x = 2$
 (D) none of these

यदि $f(x) = \begin{cases} x + \{x\} + x \sin \{x\} & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$ जहाँ $\{x\}$ एक भिन्नांक फलन हो, तो :

- (A) $x = 0$ पर 'f' सतत् तथा अवकलनीय दोनों होगा ।
 (B) $x = 0$ पर 'f' सतत् होगा लेकिन अवकलनीय नहीं होगा ।
 (C) $x = 2$ पर फलन 'f' सतत् तथा अवकलनीय दोनों होगा ।
 (D) इनमें से कोई नहीं

Ans. D

Sol. $f(0) = \lim_{x \rightarrow 0} f(x) = 0 - 1 + 0 \cdot \sin(-1) = -1$

$$f(0^+) = \lim_{x \rightarrow 0} f(x) = 0 + 0 + 0 \cdot \sin 0 = 0 = f(0)$$

$f(x)$ is not continuous at $x = 0$ at $x = 2$,

$$f(2^+) = 2 + 2 + 2 \sin 2 = 4 + 2 \sin 2$$

$$f(2^-) = 2 + 1 + 2 \sin 1 = 3 + 2 \sin 1$$

$f(x)$ is not continuous at $x = 2$

13. The set of all points where the function $f(x) = \frac{x}{1+|x|}$ is differentiable is :

- (A) $(-\infty, \infty)$ (B) $[0, \infty)$ (C) $(-\infty, 0) \cup (0, \infty)$ (D) $(0, \infty)$ (E) none

ऐसे सारे बिन्दुओं का समुच्चय, जहाँ $f(x) = \frac{x}{1+|x|}$ अवकलनीय होगा :

- (A) $(-\infty, \infty)$ (B) $[0, \infty)$ (C) $(-\infty, 0) \cup (0, \infty)$ (D) $(0, \infty)$ (E) इनमें से कोई नहीं

Ans. A

Sol. $f(x) = \begin{cases} \frac{x}{1-x} & , x < 0 \\ \frac{x}{1+x} & , x \geq 0 \end{cases}$

$f(x)$ is continuous and differentiable for $x \in \mathbb{R}$

14. Consider the function $f(x) = \begin{cases} x\{x\} + 1 & 0 \leq x < 1 \\ 2 - \{x\} & 1 \leq x \leq 2 \end{cases}$ where $\{x\}$ denotes the fractional part function. Which one of the following statements is NOT correct :

- (A) $\lim_{x \rightarrow 1} f(x)$ exists (B) $f(0) \neq f(2)$
(C) $f(x)$ is continuous in $[0, 2]$ (D) None

माना कि $f(x) = \begin{cases} x\{x\} + 1 & 0 \leq x < 1 \\ 2 - \{x\} & 1 \leq x \leq 2 \end{cases}$ जहाँ $\{x\}$ एक भिन्नात्मक फलन हो, तो निम्न में से कौन सा विकल्प सही नहीं है :

- (A) $\lim_{x \rightarrow 1} f(x)$ का अस्तित्व है। (B) $f(0) \neq f(2)$
(C) $[0, 2]$ के अन्तराल में $f(x)$ सतत् है। (D) इनमें से कोई नहीं

Ans. C

Sol. $f(1^+) = f(1^-) = f(1) = 2$
 $f(0) = 1, \quad f(2) = 2$
 $f(2^-) = 1; \quad f(2) = 2 \Rightarrow f$ is not continuous at $x = 2$

15. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $g(2) = -40$ and $g'(2) = -5$.

Then $\lim_{x \rightarrow 0} \left(\frac{g(2-x^2)}{g(2)} \right)^{\frac{4}{x^2}}$ is equal to :

माना $g : \mathbb{R} \rightarrow \mathbb{R}$ एक अवकलनीय फलन इस प्रकार है कि $g(2) = -40$ तथा $g'(2) = -5$.

तो $\lim_{x \rightarrow 0} \left(\frac{g(2-x^2)}{g(2)} \right)^{\frac{4}{x^2}}$ का मान होगा :

- (A) e^{32} (B) $\sqrt{e}$ (C) $\frac{1}{\sqrt{e}}$ (D) e^{-5}

Ans. C

Sol. $\lim_{x \rightarrow 0} \left(\frac{g(2-x^2)}{g(2)} \right)^{\frac{4}{x^2}}$ (1^∞ form) $= e^{\lim_{x \rightarrow 0} 4 \left(\frac{g(2-x^2)-g(2)}{x^2} \right) \times \frac{1}{g(2)}} = e^{-4 \frac{g'(2)}{g(2)}} = e^{\frac{-4 \times (-5)}{-40}} = e^{\frac{-1}{2}} = \frac{1}{\sqrt{e}}$.

16. Let $f : (0, 1) \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{b-x}{1-bx}$, where b is a constant such that $0 < b < 1$. Then : [JEE - 2011]

- (A) f is not invertible on $(0, 1)$ (B) $f \neq f^{-1}$ on $(0, 1)$ and $f(b) = \frac{1}{f'(0)}$
(C) $f = f^{-1}$ on $(0, 1)$ and $f(b) = \frac{1}{f'(0)}$ (D) f^{-1} is differentiable on $(0, 1)$

माना $f: (0, 1) \rightarrow \mathbb{R}$ इस प्रकार परिभाषित है $f(x) = \frac{b-x}{1-bx}$, जहाँ b एक स्थिरांक है इस प्रकार कि $0 < b < 1$ तो :

[JEE - 2011]

- (A) f , अन्तराल $(0, 1)$ में व्युत्क्रमणीय नहीं है (B) $f \neq f^{-1}$ on $(0, 1)$ तथा $f(b) = \frac{1}{f'(0)}$
- (C) $f = f^{-1}$ on $(0, 1)$ तथा $f(b) = \frac{1}{f'(0)}$ (D) f^{-1} , अन्तराल $(0, 1)$ में अवकलनीय है

Ans. A

Sol. $f: (0,1) \rightarrow \mathbb{R}$

$$f(x) = \frac{b-x}{1-bx} \quad b \in (0,1)$$

$$\Rightarrow f'(x) = \frac{b^2-1}{(bx-1)^2}$$

$$\Rightarrow f'(x) < 0 \quad \forall x \in (0, 1)$$

hence $f(x)$ is decreasing function

hence its range $(-1, b)$

$\Rightarrow$ co-domain $\neq$ range

$\Rightarrow f(x)$ is non-invertible function

17. Let $S = \{t \in \mathbb{R}: f(x) = |x - \pi| \cdot (e^{|x|} - 1) \sin |x| \text{ is not differentiable at } t\}$. Then the set S is equal to :

[JEE (Main) 2018]

- (A) $\{\pi\}$ (B) $\{0, \pi\}$ (C) ϕ (an empty set) (D) $\{0\}$

माना $S = \{t \in \mathbb{R}: f(x) = |x - \pi| \cdot (e^{|x|} - 1) \sin |x| \text{ जो पर अवकलनीय नहीं है}\}$, तो समुच्चय S बराबर है :

[JEE (Main) 2018]

- (A) $\{\pi\}$ (B) $\{0, \pi\}$ (C) ϕ (एक रिक्त समुच्चय) (D) $\{0\}$

Ans. C

Sol. We will check at $x = 0, \pi$

$$f(x) = \begin{cases} (x - \pi)(e^x - 1)\sin x & x \geq \pi \\ (\pi - x)(e^x - 1)\sin x & 0 \leq x < \pi \\ -(\pi - x)(e^x - 1)\sin x & x < 0 \end{cases}$$

at $x = 0$

$$f(0^+) = 0$$

$$f(0^-) = 0$$

at $x = \pi$

$$f(\pi^+) = 0$$

$$f(\pi^-) = 0$$

It is differentiable at $x = 0, \pi$

$$S = \{ \} = \phi$$

DIFFERENTIABILITY CHECK

18. Let f be an injective and differentiable function such that $f(x) \cdot f(y) + 2 = f(x) + f(y) + f(xy)$ for all non negative real x and y with $f'(0) = 0, f'(1) = 2 \neq f(0)$, then :

$$(A) \quad x f'(x) - 2 f(x) + 2 = 0$$

$$(B) \quad x f'(x) + 2 f(x) - 2 = 0$$

$$(C) \quad x f'(x) - f(x) + 1 = 0$$

$$(D) \quad 2 f(x) = f'(x) + 2$$

f एक एकैक तथा अवकलनीय फलन हो तथा यदि $f(x) \cdot f(y) + 2 = f(x) + f(y) + f(xy) \quad \forall x, y \in \mathbb{R}^+ \cup$

$\{0\}$ तथा $f'(0) = 0, f'(1) = 2 \neq f(0)$, तो :

Ans. A

Sol. $f(x) \cdot f(y) + 2 = f(x) + f(y) + f(xy); \quad x = 0, y = 0 \quad \Rightarrow f(0) = 1$

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f\left(x\left(1+\frac{h}{x}\right)\right) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x) \cdot f\left(1+\frac{h}{x}\right) + 2 - 2f(x) - f\left(1+\frac{h}{x}\right)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\left(f\left(1+\frac{h}{x}\right) - 2\right)}{h} (f(x) - 1) \quad \{\because f(x) \text{ be an injective} \Rightarrow f(1) = 2\}$$

$$= \frac{1}{x} \lim_{h \rightarrow 0} \left(\frac{f\left(1+\frac{h}{x}\right) - f(1)}{\frac{h}{x}} \right) (f(x) - 1)$$

$$f'(x) = \frac{1}{x} f'(1) (f(x) - 1) \Rightarrow x f'(x) = 2 f(x) - 2$$

19. Let $f(x)$ be continuous and differentiable function for all reals.

$f(x+y) = f(x) - 3xy + f(y)$. If $\lim_{h \rightarrow 0} \frac{f(h)}{h} = 7$, then the value of $f'(x)$ is :

फलन $f(x)$, x के प्रत्येक मान के लिये सतत् तथा अवकलनीय हो तथा

$f(x+y) = f(x) - 3xy + f(y)$. यदि $\lim_{h \rightarrow 0} \frac{f(h)}{h} = 7$, तो $f'(x)$ का मान होगा :

- (A) $-3x$ (B) 7 (C) $-3x + 7$ (D) $2f(x) + 7$

Ans. C

Sol. $f'(x) = \frac{f(x+h) - f(x)}{h} = \frac{f(h) - 3xh}{h} = 7 - 3x$

20. If $f(x+y) = f(x) + f(y) + c$, for all real x and y and $f(x)$ is continuous at $x = 0$ and $f'(0) = 1$ then $f'(x)$ equals to:
यदि $f(x+y) = f(x) + f(y) + c$, $\forall x, y \in \mathbb{R}$ और $x = 0$ पर $f(x)$ एक सतत् फलन है तथा $f'(0) = 1$ तो $f'(x)$ का मान होगा :

- (A) c (B) -1 (C) 0 (D) 1

Ans. D

Sol. $f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = 1$ also $f(0) = -c$

$$\begin{aligned} f'(x) &= f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x) + f(h) + c - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = f'(0) = 1 \end{aligned}$$

$$\therefore f'(x) = 1$$

21. If for all values of x & y , $f(x+y) = f(x) \cdot f(y)$ and $f(5) = 2$, $f'(0) = 3$, then $f'(5)$ is :
यदि x तथा y के सभी मानों के लिये $f(x+y) = f(x) \cdot f(y)$ तथा $f(5) = 2$, $f'(0) = 3$, तो $f'(5)$ है :

- (A) 3 (B) 4 (C) 5 (D) 6

Ans. D

Sol. $f(x+y) = f(x) \cdot f(y) \quad \forall x, y$

$$\therefore f(5+0) = f(5) \cdot f(0) \quad \{ \because f(5) = 2$$

$$\therefore f(0) = 1$$

$$\text{Now } f'(5) = \lim_{h \rightarrow 0} \frac{f(5+h) - f(5)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(5)f(h) - f(5)}{h}$$

$$= f(5) \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$$

$$= f(5) f'(0) = 2 \times 3 = 6$$

22. If f is a real-valued differentiable function satisfying $|f(x) - f(y)| \leq (x - y)^2$, $x, y \in \mathbb{R}$ and $f(0) = 0$, then $f(1)$ equals
यदि f एक वास्तविक फलन है, जो अवकलनीय है तथा $|f(x) - f(y)| \leq (x - y)^2$, $x, y \in \mathbb{R}$ को सन्तुष्ट करता है तथा $f(0) = 0$ है, तो $f(1)$ का मान है :

(A) -1

(B) 0

(C) 2

(D) 1

Ans. B

Sol. $|f(x) - f(y)| \leq |x - y|^2$

$$\Rightarrow \frac{|f(x) - f(y)|}{|x - y|} \leq |x - y|$$

$$\Rightarrow \lim_{x \rightarrow y} \left| \frac{f(x) - f(y)}{x - y} \right| \leq \lim_{x \rightarrow y} |x - y|$$

$$\Rightarrow f'(x) \leq 0 \Rightarrow f'(x) = 0$$

$\Rightarrow f(x)$ is continuous function

$$\therefore f(1) = 0 = f(0)$$

23. Suppose $f(x)$ is differentiable at $x = 1$ and $\lim_{h \rightarrow 0} \frac{1}{h} f(1 + h) = 5$, then $f'(1)$ equals :

माना $f(x)$ बिन्दु $x = 1$ पर अवकलनीय है तथा $\lim_{h \rightarrow 0} \frac{1}{h} f(1 + h) = 5$ तो $f'(1)$ का मान है :

(A) 3

(B) 4

(C) 5

(D) 6

Ans. C

Sol. Apply L Hospital rule

$$\lim_{h \rightarrow 0} \frac{f'(1 + h)}{1} = 5$$

$$\Rightarrow f'(1) = 5$$

CONTINUITY & DIFFERENTIABILITY

JEE EXERCISES

Exercise J-2

CONTINUITY CHECK

1. If the function $f(x) = \frac{3x^2 + ax + a + 3}{x^2 + x - 2}$ is continuous at $x = -2$. Find $f(-2)$:

यदि फलन $f(x) = \frac{3x^2 + ax + a + 3}{x^2 + x - 2}$, $x = -2$ पर सतत् हो, तो $f(-2)$ का मान ज्ञात करो :

Ans. -1

Sol. $f(x) = \frac{3x^2 + ax + a + 3}{x^2 + x - 2}$ is continuous at $x = -2$

$$\Rightarrow \lim_{x \rightarrow -2} f(x) = f(-2)$$

$$\Rightarrow \lim_{x \rightarrow -2} \frac{3x^2 + a(1+x) + 3}{x^2 + x - 2} \left(\frac{0}{0} \text{ form} \right) \Rightarrow a = 15$$

$$= \lim_{x \rightarrow -2} \frac{3x^2 + 15x + 18}{(x+2)(x-1)}$$

$$= 3 \lim_{x \rightarrow -2} \frac{(x+2)(x+3)}{(x+2)(x-1)} = \frac{3 \cdot 1}{-3} = -1$$

$$\therefore f(-2) = -1$$

2. Find all possible values of a and b so that $f(x)$ is continuous for all $x \in \mathbb{R}$ if

$$f(x) = \begin{cases} |ax+3| & \text{if } x \leq -1 \\ |3x+a| & \text{if } -1 < x \leq 0 \\ \frac{b \sin 2x}{x} - 2b & \text{if } 0 < x < \pi \\ \cos^2 x - 3 & \text{if } x \geq \pi \end{cases} :$$

यदि फलन $f(x)$, x के प्रत्येक वास्तविक मान के लिये सतत् हो, जहाँ

$$f(x) = \begin{cases} |ax+3| & \text{if } x \leq -1 \\ |3x+a| & \text{if } -1 < x \leq 0 \\ \frac{b \sin 2x}{x} - 2b & \text{if } 0 < x < \pi \\ \cos^2 x - 3 & \text{if } x \geq \pi \end{cases} \quad \text{तो } a \text{ व } b \text{ के सभी सम्भव मान ज्ञात करो :}$$

Ans. $a = 0, b = 1$

Sol.
$$f(x) = \begin{cases} |ax+3| & x \leq 1 \\ |3x+a| & -1 \leq x \leq 0 \\ \frac{b \sin 2x}{x} - 2b & \text{if } 0 < x < \pi \\ \cos^2 x - 3 & \text{if } x \geq \pi \end{cases}$$

Given $f(x)$ is continuous $\forall x \in \mathbb{R}$

At $x = 0$

$$\text{L.H.L.} = |a| = f(0)$$

$$\text{R.H.L.} = \lim_{h \rightarrow 0} \left(\frac{b \sin 2h}{h} - 2b \right) = 0$$

$$\Rightarrow |a| = 0$$

$$\Rightarrow a = 0$$

at $x = +\pi$

$$f(\pi^+) = f(\pi) = -2$$

$$\text{L.H.L.} = \lim_{h \rightarrow 0} \frac{\sin(2\pi - 2h)}{\pi - h} - 2b = -2b$$

$$\Rightarrow b = 1$$

3. The function $f(x) = \begin{cases} \left(\frac{6}{5}\right)^{\frac{\tan 6x}{\tan 5x}} & \text{if } 0 < x < \frac{\pi}{2} \\ b+2 & \text{if } x = \frac{\pi}{2} \\ (1+|\cos x|)^{\left(\frac{a|\tan x|}{b}\right)} & \text{if } \frac{\pi}{2} < x < \pi \end{cases}$

Determine the values of 'a' & 'b', if f is continuous at $x = \pi/2$:

दिया गया फलन $f(x) = \begin{cases} \left(\frac{6}{5}\right)^{\frac{\tan 6x}{\tan 5x}} & \text{if } 0 < x < \frac{\pi}{2} \\ b+2 & \text{if } x = \frac{\pi}{2} \\ (1+|\cos x|)^{\left(\frac{a|\tan x|}{b}\right)} & \text{if } \frac{\pi}{2} < x < \pi \end{cases}$

यदि फलन $x = \pi/2$ पर सतत् हो, तो 'a' व 'b' का मान ज्ञात करो :

Ans $a = 0 ; b = -1$

Sol.

$$f(x) = \begin{cases} \left(\frac{6}{5}\right)^{\frac{\tan 6x}{\tan 5x}} & \text{if } 0 < x < \frac{\pi}{2} \\ b + 2 & \text{if } x = \frac{\pi}{2} \\ (1 + |\cos x|)^{\left(\frac{a|\tan x|}{b}\right)} & \text{if } \frac{\pi}{2} < x < \pi \end{cases}$$

Given f is continuous at $x = \frac{\pi}{2}$

$$\text{L.H.L.} = \lim_{h \rightarrow 0} f\left(\frac{\pi}{2} - h\right)$$

$$= \lim_{h \rightarrow 0} \left(\frac{6}{5}\right)^{\frac{\tan\left(\frac{6\pi}{2} - 6h\right)}{\tan\left(\frac{5\pi}{2} - 5h\right)}}$$

$$= \lim_{h \rightarrow 0} \left(\frac{6}{5}\right)^{-\frac{\tan 6h}{\cot 5h}}$$

$$= \lim_{h \rightarrow 0} \left(\frac{6}{5}\right)^{-\tan 5h \cdot \tan 6h} = 1$$

$$\Rightarrow b + 2 = 1$$

$$\Rightarrow b = -1$$

$$\text{R.H.L.} = \lim_{h \rightarrow 0} f\left(\frac{\pi}{2} + h\right) = \lim_{h \rightarrow 0} (1 + |-\sinh|)^{\left|\frac{a \coth}{b}\right|}$$

$$= \lim_{h \rightarrow 0} (1 - \sinh)^{a \coth} = 1$$

$$\text{if } a = 0$$

4. Suppose that $f(x) = x^3 - 3x^2 - 4x + 12$ and $h(x) = \begin{cases} \frac{f(x)}{x-3} & , x \neq 3 \\ K & , x = 3 \end{cases}$ then

(a) find all zeros of $f(x)$

(b) find the value of K that makes h continuous at $x = 3$

(c) using the value of K found in (b), determine whether h is an even function :

माना कि $f(x) = x^3 - 3x^2 - 4x + 12$ और $h(x) = \begin{cases} \frac{f(x)}{x-3} & , x \neq 3 \\ K & , x = 3 \end{cases}$ तो,

(a) $f(x) = 0$ के मूल बताइये

(b) यदि फलन $h(x)$, $x = 3$ पर सतत् हो, तो K का मान बताओ।

(c) (b) में प्राप्त K के मान का उपयोग करते हुए बताइयें कि क्या $h(x)$ एक सम फलन होगा :

Ans. (a) -2, 2, 3 (b) $K = 5$ (c) even

Sol. $f(x) = (x+2)(x-2)(x-3)$

$$h(x) = \begin{cases} (x+2)(x-2) & , x \neq 3 \\ k & , x = 3 \end{cases} \quad \text{for continuity} \quad k = \lim_{x \rightarrow 3} h(x) = 5$$

$$h(x) = (x+2)(x-2) = x^2 - 4 \text{ which is even } \forall x \in \mathbb{R}$$

5. Let $f(x) = \begin{cases} \frac{1-\sin \pi x}{1+\cos 2\pi x} & , x < \frac{1}{2} \\ p & , x = \frac{1}{2} \\ \frac{\sqrt{2x-1}}{\sqrt{4+\sqrt{2x-1}}-2} & , x > \frac{1}{2} \end{cases}$. Determine the value of p , if possible, so that the function is continuous at $x=1/2$:

माना कि $f(x) = \begin{cases} \frac{1-\sin \pi x}{1+\cos 2\pi x} & , x < \frac{1}{2} \\ p & , x = \frac{1}{2} \\ \frac{\sqrt{2x-1}}{\sqrt{4+\sqrt{2x-1}}-2} & , x > \frac{1}{2} \end{cases}$. यदि फलन $x=1/2$ पर सतत् हो, तो यदि संभव है तो p का मान बताइये :

Ans p not possible

Sol. $f\left(\frac{1}{2}\right) = P$

$$f\left(\frac{1}{2}^+\right) = \lim_{h \rightarrow 0} f\left(\frac{1}{2} + h\right)$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{2h}}{\sqrt{4+\sqrt{2h}}-2} = \lim_{h \rightarrow 0} \frac{\sqrt{2h}(\sqrt{4+\sqrt{2h}}+2)}{2+\sqrt{2h}} = 0$$

$$f\left(\frac{1}{2}^-\right) = \lim_{h \rightarrow 0} f\left(\frac{1}{2} - h\right)$$

$$= \lim_{h \rightarrow 0} \frac{1 - \sin\left(\frac{\pi}{2} - \pi h\right)}{1 + \cos(\pi - 2\pi h)}$$

$$= \lim_{h \rightarrow 0} \frac{1 - \cos \pi h}{1 - \cos(2\pi h)} = \frac{2 \sin^2 \frac{\pi}{h}}{2 \sin^2 \pi h} \cdot \frac{\pi^2 h^2}{4 \cdot \frac{\pi^2 h^2}{4}}$$

$$= \frac{1}{4}$$

$\Rightarrow f(x)$ is not continuous at $x = \frac{1}{2}$.

No value of P is possible for which $f(x)$ is continuous.

6. Given the function $g(x) = \sqrt{6-2x}$ and $h(x) = 2x^2 - 3x + a$. Then :

(a) evaluate $h(g(2))$ (b) If $f(x) = \begin{cases} g(x), & x \leq 1 \\ h(x), & x > 1 \end{cases}$, find 'a' so that f is continuous.

फलन $g(x) = \sqrt{6-2x}$ तथा $h(x) = 2x^2 - 3x + a$. दिया गया है

(a) $h(g(2))$ का मान बताओ। (b) यदि $f(x) = \begin{cases} g(x), & x \leq 1 \\ h(x), & x > 1 \end{cases}$, तो 'a' का मान बताओ जबकि फलन f सतत् हो।

Ans. (a) $4 - 3\sqrt{2} + a$, (b) $a = 3$

Sol. (a) $h(g(2)) = h(\sqrt{2}) = 2 \cdot 2 - 3\sqrt{2} + a$
 $= 4 - 3\sqrt{2} + a$

$$(b) f(x) = \begin{cases} \sqrt{6-2x} & x \leq 1 \\ 2x^2 - 3x + a & x > 1 \end{cases}$$

If f is continuous at $x = 1$ then $f(1) = \sqrt{4} = 2$ & $f(1^+) = 2 - 3 + a = a - 1$

$$\Rightarrow a = 2 + 1 = 3$$

7. Determine the values of a, b & c for which the function $f(x) = \begin{cases} \frac{\sin(a+1)x + \sin x}{x} & \text{for } x < 0 \\ c & \text{for } x = 0 \\ \frac{(x+bx^2)^{1/2} - x^{1/2}}{bx^{3/2}} & \text{for } x > 0 \end{cases}$

is continuous at $x = 0$:

यदि फलन $f(x) = \begin{cases} \frac{\sin(a+1)x + \sin x}{x} & \text{for } x < 0 \\ c & \text{for } x = 0 \\ \frac{(x+bx^2)^{1/2} - x^{1/2}}{bx^{3/2}} & \text{for } x > 0 \end{cases}$, $x = 0$ पर सतत् हो, तो a, b व c के मान ज्ञात करो :

Ans. $a = -3/2$, $b \neq 0$, $c = 1/2$

Sol. $\lim_{x \rightarrow 0^-} \frac{\sin(a+1)x + \sin x}{x} = a + 2$

and $\lim_{x \rightarrow 0^+} \frac{x + bx^2 - x}{bx^{3/2}(\sqrt{x + bx^2} + \sqrt{x})} = \frac{1}{2}$ as $b \neq 0$

according to question

$$c = \frac{1}{2} \text{ \& } a + 2 = \frac{1}{2} \Rightarrow a = \frac{-3}{2}$$

8. If $f(x) = \frac{\sin 3x + A \sin 2x + B \sin x}{x^5}$ ($x \neq 0$) is cont. at $x = 0$. Find A & B. Also find $f(0)$.

यदि फलन $f(x) = \frac{\sin 3x + A \sin 2x + B \sin x}{x^5}$ ($x \neq 0$) पर सतत् हो, तो A व B का मान ज्ञात करो तथा $f(0)$ का भी मान बताओ।

Ans. $A = -4$, $B = 5$, $f(0) = 1$

Sol. $f(x) = \frac{\sin 3x + A \sin 2x + B \sin x}{x^5}; x \neq 0$

Given $f(x)$ is continuous at $x = 0$

$$\Rightarrow f(0) = \lim_{x \rightarrow 0} f(x)$$

$$= \lim_{x \rightarrow 0} \frac{\left(3x - \frac{(3x)^3}{|3|} + \dots \right) + A \left(2x - \frac{(2x)^3}{|3|} + \frac{(2x)^5}{|5|} + \dots \right) + B \left(x - \frac{x^3}{|3|} + \frac{x^5}{|5|} + \dots \right)}{x^5}$$

$$= \lim_{x \rightarrow 0} \frac{(3 + 2A + B) - \frac{x^3}{|3|}(27 + 8A + B) + \frac{x^5}{|5|}(32A + 243B) + \dots}{x^5}$$

$$\Rightarrow 3 + 2A + B = 0$$

$$27 + 8A + B = 0$$

$$\Rightarrow A = -4 \text{ \& } B = 5$$

$$\Rightarrow f(0) = \frac{32A + 243 + B}{|5|}$$

$$= \frac{120}{120} = 1$$

9. Let $f(x) = \begin{cases} \frac{\left(\frac{\pi}{2} - \sin^{-1}(1 - \{x\}^2)\right) \sin^{-1}(1 - \{x\})}{\sqrt{2}(\{x\} - \{x\}^3)} & \text{for } x \neq 0 \\ \frac{\pi}{2} & \text{for } x = 0 \end{cases}$ where $\{x\}$ is the fractional part of x .

Consider another function $g(x)$; such that $g(x) = \begin{cases} f(x) & \text{for } x \geq 0 \\ 2\sqrt{2}f(x) & \text{for } x < 0 \end{cases}$.

Discuss the continuity of the functions $f(x)$ & $g(x)$ at $x = 0$:

दिया हो कि $f(x) = \begin{cases} \frac{\left(\frac{\pi}{2} - \sin^{-1}(1 - \{x\}^2)\right) \sin^{-1}(1 - \{x\})}{\sqrt{2}(\{x\} - \{x\}^3)} & \text{for } x \neq 0 \\ \frac{\pi}{2} & \text{for } x = 0 \end{cases}$ जहाँ $\{x\}$ x का भिन्नांक फलन है एक दूसरा फलन $g(x)$ इस

प्रकार से परिभाषित है कि $g(x) = \begin{cases} f(x) & \text{for } x \geq 0 \\ 2\sqrt{2}f(x) & \text{for } x < 0 \end{cases}$

तो $f(x)$ तथा $g(x)$ की $x = 0$ पर सतता की जाँच करिये :

Ans. $f(0^+) = \frac{\pi}{2}$; $f(0^-) = \frac{\pi}{4\sqrt{2}} \Rightarrow f$ is discontin. at $x = 0$; $g(0^+) = g(0^-) = g(0) = \pi/2 \Rightarrow g$ is cont. at $x = 0$

Sol. $\lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} \frac{\left(\frac{\pi}{2} - \sin^{-1}(1 - \{h\}^2)\right) \sin^{-1}(1 - \{h\})}{\sqrt{2}(\{h\} - \{h\}^3)}$

$$= \lim_{h \rightarrow 0} \frac{\left(\frac{\pi}{2} - \sin^{-1}(1 - h^2)\right) \sin^{-1}(1 - h)}{\sqrt{2}(h - h^3)}$$

$$= \lim_{h \rightarrow 0} \frac{\cos^{-1}(1 - h^2)}{\sqrt{2}(1 - h^2)} \times \frac{\sin^{-1}(1 - h)}{h} = \frac{\pi}{2}$$

$$\lim_{h \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} \frac{\left(\frac{\pi}{2} - \sin^{-1}(1 - (1 - h)^2)\right) \sin^{-1}(1 - (1 - h))}{\sqrt{2}((1 - h) - (1 - h)^3)}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{\pi}{2} \sin^{-1} h}{\sqrt{2}(1 - h)(2 - h)h} = \frac{\pi}{4\sqrt{2}}$$

so $f(x)$ is discontinuous at $x = 0$

$$\text{Now } g(x) = \begin{cases} \frac{\pi}{2} & ; x \geq 0 \\ 2\sqrt{2} \frac{\pi}{4\sqrt{2}} & ; x < 0 \end{cases}$$

$$g(x) = \begin{cases} \frac{\pi}{2} & ; x \geq 0 \\ \frac{\pi}{2} & ; x < 0 \end{cases}$$

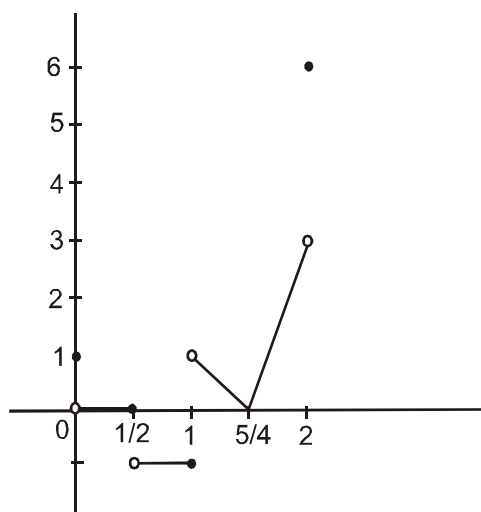
so $g(x)$ is continuous at $x = 0$

10. Discuss the continuity of f in $[0, 2]$ where $f(x) = \begin{cases} 4x - 5[x] & \text{for } x > 1 \\ [\cos \pi x] & \text{for } x \leq 1 \end{cases}$; where $[x]$ is the greatest integer not greater than x . Also draw the graph :

यदि $f(x) = \begin{cases} 4x - 5[x] & \text{for } x > 1 \\ [\cos \pi x] & \text{for } x \leq 1 \end{cases}$ जहाँ $[x]$ महत्तम पूर्णांक फलन हो, तो f की $[0, 2]$ अन्तराल में सतता की जाँच करिये तथा $y = f(x)$ का लेखा चित्र भी खींचियें :

Ans. the function f is continuous everywhere in $[0, 2]$ except for $x = 0, \frac{1}{2}, 1$ & 2 .

Sol. $f(x) = \begin{cases} 1 & , x = 0 \\ 0 & , 0 < x \leq 1/2 \\ -1 & , 1/2 < x \leq 1 \\ 5 - 4x & , 1 < x < 5/4 \\ 4x - 5 & , 5/4 \leq x < 2 \\ 6 & , x = 2 \end{cases}$



$f(x)$ (असत्त है) is discontinuous at $x = 0, 1/2, 1, 2$ in $[0, 2]$

11. If $f(x) = x + \{-x\} + [x]$, where $[x]$ is the integral part & $\{x\}$ is the fractional part of x . Discuss the continuity of f in $[-2, 2]$:

यदि $f(x) = x + \{-x\} + [x]$, जहाँ $[x]$ महत्तमपूर्णांक फलन तथा $\{x\}$ भिन्नांश फलन हो, तो $f(x)$ की $[-2, 2]$ के अन्तराल में सतता की जाँच करिये :

Ans. discontinuous at all integral values in $[-2, 2]$

Sol. $f(x) = x + \{-x\} + [x]$
 $= x - x - [-x] + [x]$
 $= [x] - [-x]$

$$f(x) = \begin{cases} -4 & x = -2 \\ -3 & -2 < x < -1 \\ -2 & x = -1 \\ -2 & -1 < x < 0 \\ -1 & x = 0 \\ 0 & 0 < x < 1 \\ 2 & x = 1 \\ 3 & 1 < x < 2 \\ 4 & x = 2 \end{cases}$$

$\Rightarrow f(x)$ is discontinuous at $x = -2, -1, 0, 1, 2$.

12. Find the locus of (a, b) for which the function $f(x) = \begin{cases} ax - b & \text{for } x \leq 1 \\ 3x & \text{for } 1 < x < 2 \\ bx^2 - a & \text{for } x \geq 2 \end{cases}$

is continuous at $x = 1$ but discontinuous at $x = 2$:

यदि फलन $f(x) = \begin{cases} ax - b & \text{for } x \leq 1 \\ 3x & \text{for } 1 < x < 2 \\ bx^2 - a & \text{for } x \geq 2 \end{cases}$, $x = 1$ पर सतत् तथा $x = 2$ पर असतत् हो, तो बिन्दु (a, b) का बिन्दुपथ ज्ञात

करिये :

Ans. locus $(a, b) \rightarrow x, y$ is $y = x - 3$ excluding the points where $y = 3$ intersects it

Sol. $f(x) = \begin{cases} ax - b & x \leq 1 \\ 3x & 1 < x < 2 \\ bx^2 - a & x \geq 2 \end{cases}$

at $x = 1$

$$f(1) = a - b$$

at $x = 2$

$$f(2^-) = 6$$

$$f(1^+) = 3$$

$$f(1^-) = a - b$$

$$\Rightarrow a - b = 3$$

$$f(2^+) = f(2) = 4b - a$$

$$\Rightarrow 4b - a \neq 6$$

$$\Rightarrow 4b - (3 + b) \neq 6$$

$$\Rightarrow 3b \neq 9$$

$$\Rightarrow b \neq 3$$

Locus of (a, b) is

$y = x - 3$; excluding the points where $y = 3$ intersects it.

13. A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined as $f(x) = \lim_{n \rightarrow \infty} \frac{ax^2 + bx + c + e^{nx}}{1 + c \cdot e^{nx}}$ where f is continuous on $\mathbb{R}$. Find the values of a , b and c :

एक फलन $f: \mathbb{R} \rightarrow \mathbb{R}$ इस प्रकार से परिभाषित होता है कि $f(x) = \lim_{n \rightarrow \infty} \frac{ax^2 + bx + c + e^{nx}}{1 + c \cdot e^{nx}}$ यदि $f(x)$ सभी वास्तविक संख्या के

लिये सतत् हो, तो a , b तथा c का मान ज्ञात करो :

Ans. $c = 1, a, b \in \mathbb{R}$

Sol.
$$f(x) = \lim_{n \rightarrow \infty} \frac{ax^2 + bx + c + e^{nx}}{1 + ce^{nx}}$$

$$= \lim_{n \rightarrow \infty} \frac{ax^2 + bx + c + 1 + nx + \frac{n^2 x^2}{2} + \dots}{1 + c \left(1 + nx + \frac{n^2 x^2}{2} + \dots \right)}$$

$$= \lim_{n \rightarrow \infty} \frac{ax^2 + bx + c + 1 + nx + \frac{n^2 x^2}{2} + \dots}{1 + c + ncx + \frac{n^2 x^2}{2} + \dots}$$

14. Let $f(x) = x^3 - x^2 - 3x - 1$ and $h(x) = \frac{f(x)}{g(x)}$ where h is a rational function such that

(a) it is continuous every where except when $x = -1$, (b) $\lim_{x \rightarrow \infty} h(x) = \infty$ and (c) $\lim_{x \rightarrow -1} h(x) = \frac{1}{2}$.

Find $\lim_{x \rightarrow 0} (3h(x) + f(x) - 2g(x))$:

यदि $f(x) = x^3 - x^2 - 3x - 1$ तथा $h(x) = \frac{f(x)}{g(x)}$. जहाँ पर $h(x)$ एक परिमेय फलन हो तथा निम्नलिखित बातें दी गयी हैं

(a) $x = -1$ के अतिरिक्त सभी बिन्दुओं पर सतत् है (b) $\lim_{x \rightarrow \infty} h(x) = \infty$ तथा (c) $\lim_{x \rightarrow -1} h(x) = \frac{1}{2}$.

$\lim_{x \rightarrow 0} (3h(x) + f(x) - 2g(x))$ का मान बताओ :

Ans. $g(x) = 4(x+1)$ and limit $= -\frac{39}{4}$

Sol. $h(x) = \frac{x^3 - x^2 - 3x - 1}{g(x)}$

$$= \frac{(x+1)(x^2 - 2x - 1)}{g(x)}$$

$$= \frac{(x+1)(x^2 - 2x - 1)}{g(x)}$$

$$\lim_{x \rightarrow 1^-} h(x) = \frac{1}{2}$$

$$\Rightarrow \lim_{x \rightarrow 1^-} \frac{(x+1)(x^2 - 2x - 1)}{(x+1)Q(x)} = \frac{1}{2}$$

$$\Rightarrow \frac{+2}{Q(-1)} = \frac{1}{2}$$

$$\Rightarrow Q(-1) = 4$$

$$\therefore g(x) = (x+1)Q(x)$$

$$\lim_{x \rightarrow \infty} h(x) = \infty$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{x^2 - 2x - 1}{Q(x)} = \infty$$

$$\therefore Q(x) = ax + b$$

$$\Rightarrow b - a = 4$$

15. Find the number of points of discontinuity of the function $f(x) = [5x] + \{3x\}$ in $[0, 5]$. where $[y]$ and $\{y\}$ denote largest integer less than or equal to y and fractional part of y respectively :

फलन $f(x) = [5x] + \{3x\}$ के $[0, 5]$ में असतत् बिन्दुओं की संख्या क्या होगी जहाँ $[y]$ तथा $\{y\}$ महत्तम पूर्णांक फलन एवं भिन्नांश फलन को क्रमशः निरूपित करते हैं :

Ans. 30

Sol. $f(x) = [5x] - [3x] + 3x$

Now in $[0, 1]$ we have to check at $x = \frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}, \frac{1}{3}, \frac{2}{3}$

and $f(x)$ is discontinuous at $x = \frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}, \frac{1}{3}, \frac{2}{3}$ i.e. 6 points.

$\therefore$ Total points $6 \times 5 = 30$ **Ans.**

Note : $f(x)$ is continuous at all integers in $[0, 5]$

16. Let $f(x) = x^3 - 3x^2 + 6$ and $g(x) = \begin{cases} \max\{f(t), x+1 \leq t \leq x+2\} & , -3 \leq x < 0 \\ 1-x & , x \geq 0 \end{cases}$. test the continuity of $g(x)$ for

$x \in [-3, 1]$:

माना $f(x) = x^3 - 3x^2 + 6$ तथा $g(x) = \begin{cases} \max\{f(t), x+1 \leq t \leq x+2\} & , -3 \leq x < 0 \\ 1-x & , x \geq 0 \end{cases}$. तो फलन $g(x)$ की अन्तराल $x \in [-3, 1]$ में

सातत्यता की जाँच कीजिये :

Ans. Discont. at $x = 0$

Sol. $f(x) = x^3 - 3x^2 + 6$

$$g(x) = \begin{cases} \max\{f(t), x+1 \leq t \leq x+2\} & , -3 \leq x < 0 \\ 1-x & , x \geq 0 \end{cases}$$

$$= \{\max(t^3 - 3t^2 + 6; x+1 \leq t \leq x+2); -3 \leq x < 0\}$$

TYPES OF DISCONTINUITY

17. Let $f(x) = \begin{cases} \frac{\ln \cos x}{\sqrt[4]{1+x^2} - 1} & \text{if } x > 0 \\ \frac{e^{\sin 4x} - 1}{\ln(1 + \tan 2x)} & \text{if } x < 0 \end{cases}$

Is it possible to define $f(0)$ to make the function continuous at $x = 0$. If yes what is the value of $f(0)$, if not then indicate the nature of discontinuity :

$$\text{माना कि } f(x) = \begin{cases} \frac{\ln \cos x}{\sqrt[4]{1+x^2} - 1} & \text{if } x > 0 \\ \frac{e^{\sin 4x} - 1}{\ln(1 + \tan 2x)} & \text{if } x < 0 \end{cases}$$

क्या ये सम्भव है कि $f(0)$ इस प्रकार से परिभाषित हो, कि फलन $f(0)$, $x = 0$ पर सतत हो जाये? यदि हाँ, तो $f(0)$ का मान क्या होगा, यदि ना हो तो असततता का स्वभाव बताइये :

Ans. $f(0^+) = -2$; $f(0^-) = 2$ hence $f(0)$ not possible to define

Sol.

$$f(x) = \begin{cases} \frac{\ln \cos x}{\left(1+x^2\right)^{\frac{1}{4}} - 1} & \text{if } x > 0 \\ \frac{e^{\sin 4x} - 1}{\ln(1 + \tan 2x)} & \text{if } x < 0 \end{cases}$$

$$\text{L.H.L.} = f(0^-) = \lim_{h \rightarrow 0} \frac{e^{-\sin 4h} - 1}{\ln(1 - \tan 2h)}$$

$$= \lim_{h \rightarrow 0} \frac{(e^{\sin 4h} - 1) \sin 4h}{\sin 4h} \cdot \frac{\tan 2h}{e^{\sin 4h} \ln(1 - \tan 2h)} \cdot \frac{1}{\tan 2h}$$

$$= \lim_{h \rightarrow 0} \frac{2 \cdot \sin 2h \cdot \cos^2 2h}{\sin 2h} = 2$$

$$f(0^+) = \lim_{h \rightarrow 0} \frac{\ln \cosh}{(1+h^2)^{\frac{1}{4}} - 1} \times \frac{(1+h^2)^{\frac{1}{4}} + 1}{(1+h^2)^{\frac{1}{4}} + 1}$$

$$= \lim_{h \rightarrow 0} \frac{(1+h^2)^{\frac{1}{4}} + 1}{(\sqrt{1+h^2} - 1)} \ln(1 + \cosh - 1)$$

$$= \lim_{h \rightarrow 0} \frac{\left((1+h^2)^{\frac{1}{4}} + 1 \right) (\sqrt{1+h^2} + 1)}{h^2} \cdot \frac{\ln(1 + \cosh - 1)}{\cosh - 1} \cdot \cosh - 1$$

$$= \lim_{h \rightarrow 0} - \frac{4 \cdot 2 \sin^2 \frac{h}{2}}{4 \cdot \frac{h^2}{4}} = -2$$

$\therefore f(x)$ is discontinuous at $x = 0$.

INTERMEDIATE VALUE THEOREM

18. Let f be a real valued continuous function on $\mathbb{R}$ and satisfying $f(-x) - f(x) = 0 \forall x \in \mathbb{R}$. If $f(-5) = 5$, $f(-2) = 4$, $f(3) = -2$ and $f(0) = 0$ then find the minimum number of zero's of the equation $f(x) = 0$:

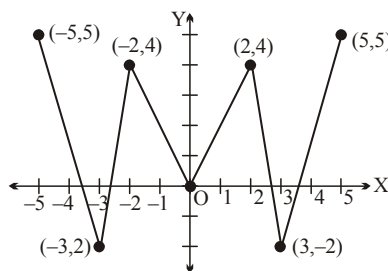
माना f , $\mathbb{R}$ में परिभाषित वास्तविक मान वाला फलन है जो $f(-x) - f(x) = 0 \forall x \in \mathbb{R}$ को संतुष्ट करता है। यदि $f(-5) = 5$, $f(-2) = 4$, $f(3) = -2$ तथा $f(0) = 0$ तो $f(x) = 0$ के न्यूनतम मूलों की संख्या बताइयें :

Ans. 5

Sol. One of the possible graph of $f(x)$

having minimum number of zeroes.

Clearly $f(x) = 0$ has atleast 5 solutions.]



19. If $g : [a, b] \rightarrow [a, b]$ is continuous show that there is some $c \in [a, b]$ such that $g(c) = c$:
 यदि $g : [a, b] \rightarrow [a, b]$ सतत् आच्छादक है तो सिद्ध करो कि कोई $c \in (a, b)$ ऐसा होगा कि $g(c) = c$:

Ans.

Sol. $g : [a, b] \xrightarrow[\text{continuous}]{\text{onto}} [a, b]$

$$f(x) = g(x) - x$$

$\underbrace{\hspace{1.5cm}}_{\substack{\text{continuous} \\ \forall x \in [a, b]}}$

$$\left. \begin{aligned} f(a) &= g(a) - a \geq 0 \\ f(b) &= g(b) - b \leq 0 \end{aligned} \right\} \because a \leq g(x) \leq b$$

By IMVT $\exists c \in (a, b)$ s.t.

$$f(c) = 0$$

$$\Rightarrow g(c) = c$$

20. Let f be continuous on the interval $[0, 1]$ to $\mathbb{R}$ such that $f(0) = f(1)$. Prove that there exists a point c in $\left[0, \frac{1}{2}\right]$ such

$$\text{that } f(c) = f\left(c + \frac{1}{2}\right) :$$

f एक $[0, 1]$ अन्तराल में एक सतत् फलन हो तथा यदि $f(0) = f(1)$ तो सिद्ध करो कि $\left[0, \frac{1}{2}\right]$ में c एक ऐसा बिन्दु होगा कि $f(c) =$

$$f\left(c + \frac{1}{2}\right) \text{ होगा :}$$

Ans.

Sol. $f : [0, 1] \xrightarrow{\text{continuous}} \mathbb{R}$

$$\text{Let } h(x) = f(x) - f\left(x + \frac{1}{2}\right)$$

$\underbrace{\hspace{1.5cm}}_{\substack{\text{continuous} \\ \text{in } \left[0, \frac{1}{2}\right]}}$

$$h(0) = f(0) - f\left(\frac{1}{2}\right)$$

$$h\left(\frac{1}{2}\right) = f\left(\frac{1}{2}\right) - f(1)$$

$$\Rightarrow h(0) + h\left(\frac{1}{2}\right) = 0$$

$\Rightarrow h(0)$ & $h\left(\frac{1}{2}\right)$ are of opposite sign.

$\therefore \exists c \in \left(0, \frac{1}{2}\right)$ s.t. $h(c) = 0$

$\Rightarrow f(c) = f\left(c + \frac{1}{2}\right)$

DIFFERENTIABILITY CHECK

21. If the function $f(x)$ defined as $f(x) = \begin{cases} -\frac{x^2}{2} & \text{for } x \leq 0 \\ x^n \sin \frac{1}{x} & \text{for } x > 0 \end{cases}$ is continuous but not derivable at $x = 0$ then find

the range of n :

यदि फलन $f(x) = \begin{cases} -\frac{x^2}{2} & \text{for } x \leq 0 \\ x^n \sin \frac{1}{x} & \text{for } x > 0 \end{cases}$, $x = 0$ पर सतत् तो है लेकिन अवकलनीय नहीं हो, तो n का परिसर ज्ञात करो :

Ans. $0 < n \leq 1$

Sol. $f(x) = \begin{cases} -\frac{x^2}{2} & x \leq 0 \\ x^n \sin \frac{1}{x} & x > 0 \end{cases}$

at $x = 0$

$f(0) = 0$

$f(0^-) = 0$

$f(0^+) = 0$ if $n > 0$

$$\text{L.H.D.} = \lim_{h \rightarrow 0} \frac{-\frac{h^2}{2} - 0}{-h} = 0$$

$$\text{R.H.D.} = \lim_{h \rightarrow 0} \frac{h^n \sin \frac{1}{h}}{h} = \lim_{h \rightarrow 0} h^{n-1} \sin \frac{1}{h} = \begin{cases} 0 & \text{if } n > 1 \\ \text{DNE} & \text{if } n \leq 1 \end{cases}$$

$\Rightarrow 0 < n \leq 1$

22. Examine the origin for continuity & derivability in the case of the function f defined by

$$f(x) = x \tan^{-1}(1/x), x \neq 0 \text{ and } f(0) = 0 :$$

यदि फलन $f(x) = x \tan^{-1}(1/x)$, $x \neq 0$ तथा $f(0) = 0$ दिया गया हो, तो $x = 0$ पर फलन की सतत्ता तथा अवकलनीयता की जाँच करो :

Ans. cont. but not diff. at $x = 0$

Sol.
$$f(x) = \begin{cases} x \tan^{-1}\left(\frac{1}{x}\right) & ; x \neq 0 \\ 0 & ; x = 0 \end{cases}$$

$$f(0^+) = \lim_{h \rightarrow 0} f(0+h) = \lim_{h \rightarrow 0} h \tan^{-1}\left(\frac{1}{h}\right) = 0$$

$$f(0^-) = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} h \tan^{-1}\left(\frac{1}{h}\right) = 0$$

$\therefore$ at $x = 0$; $f(x)$ is continuous.

$$\text{L.H.D.} = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h \rightarrow 0} \frac{+h \tan^{-1}\left(\frac{1}{h}\right)}{-h} = -\infty$$

$$\text{R.H.D.} = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = +\infty$$

23. If $f(x) = |x-1| \cdot ([x] - [-x])$, then find $f'(1^+)$ & $f'(1^-)$ where $[x]$ denotes greatest integer function :

यदि $f(x) = |x-1| \cdot ([x] - [-x])$ तो $f'(1^+)$ तथा $f'(1^-)$ का मान ज्ञात करें जहाँ $[x]$ महत्तम पूर्णांक फलन हो :

Ans. $f'(1^+) = 3$, $f'(1^-) = -1$

Sol. $f(x) = |x-1| ([x] - [-x])$

$$f'(1^+) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(1+2)}{h} = 3$$

$$f'(1^-) = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{h(0+1)}{-h} = -1$$

24. If $f(x) = \begin{cases} ax^2 - b & \text{if } |x| < 1 \\ -\frac{1}{|x|} & \text{if } |x| \geq 1 \end{cases}$ is derivable at $x = 1$. Find the values of a & b :

यदि $f(x) = \begin{cases} ax^2 - b & \text{if } |x| < 1 \\ -\frac{1}{|x|} & \text{if } |x| \geq 1 \end{cases}$, $x = 1$ पर अवकलनीय हो, तो a व b का मान ज्ञात करो :

Ans. $a = 1/2$, $b = 3/2$

Sol.
$$f(x) = \begin{cases} ax^2 - b & ; -1 < x < 1 \\ \frac{-1}{x} & ; x \geq 1 \\ \frac{1}{x} & ; x \leq -1 \end{cases}$$

Now $f(x)$ is differentiable at $x = 1$

$$\Rightarrow 2ax = \frac{1}{x^2} \quad \text{at } x = 1$$

$$\Rightarrow a = \frac{1}{2}$$

Also $f(x)$ is continuous at $x = 1$

$$\Rightarrow a(1)^2 - b = -1 \quad \Rightarrow b = \frac{3}{2}$$

25. Let $f(x)$ be defined in the interval $[-2, 2]$ such that $f(x) = \begin{cases} -1 & , -2 \leq x \leq 0 \\ x-1 & , 0 < x \leq 2 \end{cases}$ &

$g(x) = f(|x|) + |f(x)|$. Test the differentiability of $g(x)$ in $(-2, 2)$:

अन्तराल $[-2, 2]$ में फलन $f(x)$ इस प्रकार से परिभाषित है कि $f(x) = \begin{cases} -1 & , -2 \leq x \leq 0 \\ x-1 & , 0 < x \leq 2 \end{cases}$ तथा

$g(x) = f(|x|) + |f(x)|$ तो $g(x)$ की $(-2, 2)$ के अन्तराल में अवकलनीयता की जाँच करो :

Ans. not derivable at $x = 0$ & $x = 1$

Sol.
$$f(x) = \begin{cases} -1 & -2 \leq x \leq 0 \\ x-1 & 0 < x \leq 2 \end{cases}$$

$$g(x) = f(|x|) + |f(x)|$$

$$= \begin{cases} x-1+1 & -2 \leq x \leq 0 \\ x-1+1-x & 0 \leq x \leq 1 \\ 2x-2 & 1 < x \leq 2 \end{cases}$$

$$= \begin{cases} x & -2 \leq x \leq 0 \\ 0 & 0 < x \leq 1 \\ 2(x-1) & 1 < x \leq 2 \end{cases}$$

$g(x)$ is continuous $\forall x \in [-2, 2]$

$$g'(x) = \begin{cases} 1 & -2 \leq x < 0 \\ 0 & 0 < x < 1 \\ 2 & 1 < x < 2 \end{cases}$$

$g'(x)$ is not differentiable at $x = 0$ & $x = 1$ in $(-2, 2)$.

26. $f(x) = x \cdot \left(\frac{e^{[x] + |x|} - 2}{[x] + |x|} \right)$, $x \neq 0$ & $f(0) = -1$ where $[x]$ denotes greatest integer less than or equal to x . Test the

differentiability of $f(x)$ at $x = 0$:

यदि $f(x) = x \cdot \left(\frac{e^{[x] + |x|} - 2}{[x] + |x|} \right)$, $x \neq 0$ और $f(0) = -1$ जहाँ $[x]$ महत्तम पूर्णांक फलन हो, तो $f(x)$ की $x = 0$ पर अवलकनीयता की जाँच करो :

Ans. not derivable at $x = 0$

Sol.

$$f(x) = \begin{cases} x \cdot \left(\frac{e^{[x] + |x|} - 2}{[x] + |x|} \right) & x \neq 0 \\ -1 & x = 0 \end{cases}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h \cdot (e^h - 2) + h}{h^2}$$

$$= \lim_{h \rightarrow 0} \frac{e^h - 2}{h} = 1$$

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-h(e^{h-1} - 2) + h - 1}{(-1+h)h}$$

$$= \lim_{h \rightarrow 0} \frac{h(3 - e^{h-1}) - 1}{h(-1+h)} = \text{DNE}$$

$\therefore f(x)$ is not differentiable at $x = 0$.

27. Discuss the continuity & the derivability in $[0, 2]$ of $f(x) = \begin{cases} |2x-3|[x] & \text{for } x \geq 1 \\ \sin \frac{\pi x}{2} & \text{for } x < 1 \end{cases}$

where $[]$ denote greatest integer function :

फलन $f(x) = \begin{cases} |2x-3|[x] & \text{for } x \geq 1 \\ \sin \frac{\pi x}{2} & \text{for } x < 1 \end{cases}$ की $[0, 2]$ के अन्तराल में सतत्ता तथा अवकलनीयता की जाँच करो ।

जहाँ $[]$ महत्तम पूर्णांक फलन है :

Ans. f is conti. at $x = 1, 3/2$ & disconti. at $x = 2$, f is not diff. at $x = 1, 3/2, 2$

Sol.
$$f(x) = \begin{cases} |2x-3|[x] & 1 \leq x \leq 2 \\ \sin \frac{\pi x}{2} & 0 \leq x < 1 \end{cases}$$

$$= \begin{cases} 0 & x = 0 \\ \sin \frac{\pi x}{2} & 0 < x < 1 \\ |2x-3| & 1 \leq x < 2 \\ 2 & x = 2 \end{cases}$$

$$= \begin{cases} 0 & x = 0 \\ \sin \frac{\pi x}{2} & 0 < x < 1 \\ -2x+3 & 1 \leq x < \frac{3}{2} \\ 2x-3 & \frac{3}{2} \leq x < 2 \\ 2 & x = 2 \end{cases}$$

$$f(0) = 0 \quad f(0^+) = \lim_{h \rightarrow 0} \sin\left(\frac{\pi}{2}h\right) = 0$$

$$f(1^-) = \lim_{h \rightarrow 0} \sin\left(\frac{\pi}{2} - \frac{\pi}{2}h\right) = + \lim_{h \rightarrow 0} \cos\left(\frac{\pi}{2}h\right) = 1$$

$$f(1) = 1, f(1^+) = 1$$

$$f(2) = 2, f(2^-) = \lim_{h \rightarrow 0} 4 - 2h - 3 = 1$$

$$f\left(\frac{3}{2}\right) = 0, \quad f\left(\frac{3}{2}^+\right) = 0, \quad f\left(\frac{3}{2}^-\right) = 0$$

$\therefore f(x)$ is continuous at $x = 1, 0$ & $\frac{3}{2}$

discontinuous at $x = 2$ only $\rightarrow$ not differentiable at $x = 2$

$$f'(x) = \begin{cases} \cos \frac{\pi}{2} x \cdot \frac{\pi}{2} & 0 < x < 1 \\ -2 & 1 < x < \frac{3}{2} \\ 2 & \frac{3}{2} < x < 2 \end{cases}$$

$\therefore f(x)$ is not differentiable at $x = 1, \frac{3}{2}$ & 2 .

28. Let $f(x) = [3 + 4 \sin x]$ (where $[]$ denotes the greatest integer function). If sum of all the values of 'x' in $[\pi, 2\pi]$ where $f(x)$ fails to be differentiable, is $\frac{k\pi}{2}$, then find the value of k :

माना कि $f(x) = [3 + 4 \sin x]$ (जहाँ $[]$ महत्तम पूर्णांक फलन है)। यदि अन्तराल $[\pi, 2\pi]$ में ऐसे सारे 'x' का योग, जहाँ फलन अवकलनीय नहीं है, $\frac{k\pi}{2}$ है। तो k का मान बताओं :

Ans. 24

Sol. $f(x) = [3 + 4 \sin x]$

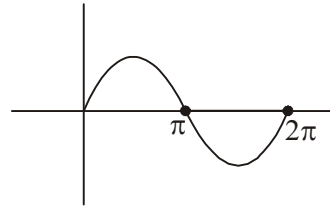
If $x \in [\pi, 2\pi]$ then $\sin x \in [-1, 0]$

$\Rightarrow 4 \sin x \in [-4, 0]$

$\Rightarrow 3 + 4 \sin x \in [-1, 3]$

$$f(x) = \begin{cases} -1 & -1 \leq 3 + 4 \sin x < 0 \\ 0 & 0 \leq 3 + 4 \sin x < 1 \\ 1 & 1 \leq 3 + 4 \sin x < 2 \\ 2 & 2 \leq 3 + 4 \sin x < 3 \\ 3 & 3 + 4 \sin x = 3 \end{cases}$$

$$f(x) = \begin{cases} -1 & -1 \leq \sin x < -\frac{3}{4} \\ 0 & -\frac{3}{4} \leq \sin x < -\frac{1}{2} \\ 1 & -\frac{1}{2} \leq \sin x < 0 \\ 2 & -\frac{1}{4} \leq \sin x < 0 \\ 3 & \sin x = 0 \end{cases}$$



clearly $f(x)$ is discontinuous at $\sin x = \frac{-3}{4}, \frac{-1}{2}, \frac{-1}{4}, 0$

$$\sin x = 0 \rightarrow x = \pi, 2\pi$$

$$\sin x = \frac{-1}{2} \rightarrow x = \pi + \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$$

$$\sin x = \frac{-1}{4} \rightarrow x = \pi + \alpha \text{ or } 2\pi - \alpha$$

$$\sin x = \frac{-3}{4}$$

$$x = \pi + \beta \text{ or } 2\pi - \beta$$

$$\Rightarrow \text{sum of all values of } x = 12\pi = \frac{24\pi}{2}$$

$$\Rightarrow k = 24$$

29. The function $f(x) = \begin{cases} ax(x-1) + b & \text{when } x < 1 \\ x-1 & \text{when } 1 \leq x \leq 3 \\ px^2 + qx + 2 & \text{when } x > 3 \end{cases}$

Find the values of the constants a, b, p, q so that

- (i) $f(x)$ is continuous for all x (ii) $f'(1)$ does not exist (iii) $f'(x)$ is continuous at $x = 3$:

दिया है कि $f(x) = \begin{cases} ax(x-1) + b & \text{when } x < 1 \\ x-1 & \text{when } 1 \leq x \leq 3 \\ px^2 + qx + 2 & \text{when } x > 3 \end{cases}$

a, b, p, q का मान ज्ञात करो, यदि

- (i) फलन $f(x)$, x के प्रत्येक मान के लिये सतत् है। (ii) $f'(1)$ विद्यमान नहीं है (iii) $x = 3$ पर $f'(x)$ सतत् हो, तो :

Ans. $a \neq 1, b = 0, p = \frac{1}{3}$ and $q = -1$

Sol. $f(x) = \begin{cases} ax^2 - ax + b & x < 1 \\ x-1 & 1 \leq x \leq 3 \\ px^2 + qx + 2 & x > 3 \end{cases}$

(i) Given $f(x)$ is continuous for all x

at $x = 1$

$$f(1^-) = b = 0 = f(1) = f(1^+)$$

$$\Rightarrow b = 0$$

at $x = 3$

$$f(3^-) = f(3) = 2 = f(3^+) = 9p + 3q + 2 \quad \dots(\text{ii})$$

Now given $f'(1)$ does not exist

$$f'(1^+) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{h-0}{h} = 1$$

$$f'(1^-) = \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{a(1-h)(1-h-1) + b}{-h}$$

$$= \lim_{h \rightarrow 0} a(1-h) = a$$

$$\therefore a \neq 1$$

$$f'(x) = \begin{cases} 2ax - a & x < 1 \\ 1 & 1 < x < 3 \\ 2px + q & x > 3 \end{cases}$$

Given $f'(x)$ is continuous at $x = 3$

$$\Rightarrow f'(3^-) = 1$$

$$f'(3^+) = 6p + q$$

$$\Rightarrow 6p + q = 1 = 6p + q \quad \dots(\text{iii})$$

$$f'(3) = \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h}$$

$$= \frac{p(3+h)^2 + 3q + qh}{h}$$

$$= \lim_{h \rightarrow 0} \frac{9p + 3q + ph^2 + 6ph + qh}{h} = 6p + q$$

By (ii) & (iii)

$$p = \frac{1}{3} \text{ \& } q = -1$$

30. Examine the function, $f(x) = x \cdot \frac{a^{1/x} - a^{-1/x}}{a^{1/x} + a^{-1/x}}$, $x \neq 0$ ($a > 0$) and $f(0) = 0$ for continuity and existence of the derivative at the origin :

फलन $f(x) = x \cdot \frac{a^{1/x} - a^{-1/x}}{a^{1/x} + a^{-1/x}}$, $x \neq 0$ ($a > 0$) और $f(0) = 0$ की $x = 0$ सतत्ता व अवकलनीयता का परिक्षण करो :

Ans. If $a \in (0, 1)$ $f'(0^+) = -1$; $f'(0^-) = 1 \Rightarrow$ continuous but not derivable

$a = 1$; $f(x) = 0$ which is constant $\Rightarrow$ continuous and derivable

If $a > 1$ $f'(0^-) = -1$; $f'(0^+) = 1 \Rightarrow$ continuous but not derivable

Sol.

$$f(x) = \begin{cases} x \cdot \frac{a^{\frac{1}{x}} - a^{-\frac{1}{x}}}{a^{\frac{1}{x}} + a^{-\frac{1}{x}}} & x \neq 0 (a > 0) \\ 0 & x = 0 \end{cases}$$

$$f(x) = \begin{cases} x \left(\frac{a^{\frac{2}{x}} - 1}{a^{\frac{2}{x}} + 1} \right) & a > 0, a \neq 1, x \neq 0 \\ 0 & a = 1, x \neq 0 \\ 0 & x = 0 \end{cases}$$

C-I $a = 1$

$\underbrace{f(x) = 0}_{\text{constant}} \rightarrow \text{cont. and derivable}$

C-II $a > 1$

$$f(0^-) = \lim_{h \rightarrow 0} \frac{-h \left(a^{\frac{2}{-h}} - 1 \right)}{a^{\frac{2}{-h}} + 1} = 0$$

$$f(0)^+ = \lim_{h \rightarrow 0} \frac{h \left(a^{\frac{2}{h}} - 1 \right)}{a^{\frac{2}{h}} + 1} = 0$$

$$f'(0^+) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{a^{\frac{2}{h}} - 1}{a^{\frac{2}{h}} + 1} = +1$$

$$f'(0^-) = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h \rightarrow 0} \left(\frac{a^{\frac{2}{-h}} - 1}{a^{\frac{2}{-h}} + 1} \right) = 1$$

$\Rightarrow f(x)$ is continuous at $x = 0$

Not differentiable at $x = 0$

C-III $0 < a < 1$

$f(0^-) = 0 = f(0^+)$

$$f'(0^-) = \lim_{h \rightarrow 0} \frac{-h \left(a^{-\frac{2}{h}} - 1 \right)}{-h \left(a^{-\frac{2}{h}} + 1 \right)} = 1$$

$$f'(0^+) = \lim_{h \rightarrow 0} \frac{h \left(a^{\frac{2}{h}} - 1 \right)}{h \left(a^{\frac{2}{h}} + 1 \right)} = -1$$

$\Rightarrow f(x)$ is continuous at $x = 0$

not differentiable at $x = 0$.

31. Discuss the continuity & the derivability of 'f' where $f(x) = \text{degree of } (u^{x^2} + u^2 + 2u - 3)$ at $x = \sqrt{2}$:

यदि $f(x) = \text{समीकरण } (u^{x^2} + u^2 + 2u - 3)$ की डिग्री है जहाँ $x = \sqrt{2}$ है, तो फलन $f(x)$ की सततता तथा अवकलनीयता की जाँच करो :

Ans. continuous but not derivable at $x = \sqrt{2}$

Sol. continuous but not derivable at $x = \sqrt{2}$

$$32. \text{ If } f(x) = \begin{cases} \frac{\sin[|x|]\pi}{a x^4} + a x^2 + b; & -2 \leq x < -1 \\ 3 \sec \pi x + \cot^{-1} x; & -1 \leq x \leq \frac{-3}{4} \end{cases}$$

is differentiable in $\left(-2, \frac{-3}{4}\right)$ then $a = \frac{1}{\lambda_1}$ and $b = \frac{3\pi - \lambda_2}{4}$. Find the value of $(\lambda_1 + \lambda_2)$.

Note: $[y]$ denotes largest integer $\leq y$:

$$\text{यदि } f(x) = \begin{cases} \frac{\sin[|x|]\pi}{a x^4} + a x^2 + b; & -2 \leq x < -1 \\ 3 \sec \pi x + \cot^{-1} x; & -1 \leq x \leq \frac{-3}{4} \end{cases}$$

Note: [y] महत्तम पूर्णांक फलन है :

$\left(-2, \frac{-3}{4}\right)$ में अवकलनीय फलन है तो $a = \frac{1}{\lambda_1}$ तथा $b = \frac{3\pi - \lambda_2}{4}$ है। $(\lambda_1 + \lambda_2)$ का मान ज्ञात कीजिये।

Ans. 17

Sol. Clearly $f(x) = \begin{cases} a x^2 + b; & -2 \leq x < -1 \\ 3 \sec \pi x + \cot^{-1} x; & -1 \leq x \leq \frac{-3}{4} \end{cases}$

As the function is differentiable in $\left(-2, \frac{-3}{4}\right)$, so function is differentiable at $x = -1$.

$$\therefore f'(-1^-) = f'(-1^+) \quad \dots(1)$$

$$\begin{aligned} f'(-1^-) &= \lim_{h \rightarrow 0} \frac{f(-1-h) - f(-1)}{-h} = \lim_{h \rightarrow 0} \frac{(a(-1-h)^2 + b) - (a + b)}{-h} \\ &= \lim_{h \rightarrow 0} \frac{a(1 + h^2 + 2h) - a}{-h} = \lim_{h \rightarrow 0} \frac{ah(h+2)}{-h} = -2a \quad \dots(2) \end{aligned}$$

$$f'(-1^+) = \lim_{h \rightarrow 0} \frac{f(-1+h) - f(-1)}{h} = \lim_{h \rightarrow 0} \frac{3 \sec \pi(-1+h) + \cot^{-1}(-1+h) - 3 \sec(-\pi) - \cot^{-1}(-1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-3 \sec \pi h + \cot^{-1}(-1+h) + 3 - \frac{3\pi}{4}}{h} = \lim_{h \rightarrow 0} \frac{3(1 - \sec \pi h)}{h} + \lim_{h \rightarrow 0} \frac{\cot^{-1}(-1+h) - \frac{3\pi}{4}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{3(\cos \pi h - 1)}{h \cos \pi h} + \lim_{\theta \rightarrow \frac{3\pi}{4}} \frac{\theta - \frac{3\pi}{4}}{1 + \cot \theta} \quad (\text{Let } \cot^{-1}(-1+h) = \theta)$$

$$= (3 \times 0) + \lim_{\theta \rightarrow 0^+} \frac{\alpha}{1 + \cot\left(\frac{3\pi}{4} + \alpha\right)} \quad \left(\text{Put } \theta = \frac{3\pi}{4} + \alpha\right)$$

$$= 0 + \lim_{\alpha \rightarrow 0} \frac{\alpha}{1 + \tan\left(\frac{\pi}{4} + \alpha\right)} = \lim_{\alpha \rightarrow 0} \frac{\alpha}{1 - \left(\frac{1 + \tan \alpha}{1 - \tan \alpha}\right)} = \lim_{\alpha \rightarrow 0} \frac{\alpha(1 - \tan \alpha)}{(-2 \tan \alpha)} = \frac{-1}{2} \quad \dots(3)$$

∴ Using equation (2) and equation (3) in equation (1), we get

$$-2a = \frac{-1}{2} \Rightarrow a = \frac{1}{4} = \frac{1}{\lambda_1} \quad (\text{Given})$$

$$\therefore \lambda_1 = 4$$

Also function will be continuous at $x = -1$.

$$\therefore \lim_{x \rightarrow -1^-} f(x) = \lim_{x \rightarrow -1^+} f(x)$$

$$\Rightarrow a + b = -3 + \frac{3\pi}{4}$$

$$\therefore b = \frac{-1}{4} - 3 + \frac{3\pi}{4} = \frac{3\pi - 13}{4} = \frac{3\pi - \lambda_2}{4}$$

$$\Rightarrow \lambda_2 = 13$$

$$\text{Hence } (\lambda_1 + \lambda_2) = 4 + 13 = 17$$

33. Let $f(x) = \begin{cases} a \cot^{-1}\left(\frac{b+x}{4}\right), & -\frac{2}{3} < x < 0 \\ 2, & x = 0 \\ \frac{\ln(1-cx)}{x}, & 0 < x < \frac{2}{3} \end{cases}$

If the function $f(x)$ is differentiable at $x = 0$, then find the value of $(b^2 - 2a + c^6)$:

माना $f(x) = \begin{cases} a \cot^{-1}\left(\frac{b+x}{4}\right), & -\frac{2}{3} < x < 0 \\ 2, & x = 0 \\ \frac{\ln(1-cx)}{x}, & 0 < x < \frac{2}{3} \end{cases}$

यदि फलन $f(x)$, $x = 0$ पर अवकलनीय है तो $(b^2 - 2a + c^6)$ का मान ज्ञात कीजिये :

Ans. 48

Sol. As $f(x)$ is derivable at $x = 0$, so $f(x)$ is also continuous at $x = 0$.

$$\therefore f(0^+) = \lim_{h \rightarrow 0} \frac{\ln(1-ch)}{h} \left(\frac{0}{0} \right) = \lim_{h \rightarrow 0} \frac{-c \times \ln(1-ch)}{-ch} = -c$$

$$\Rightarrow -c = 2 \Rightarrow c = -2 \quad \dots\dots(1)$$

$$\text{Now } f'(0^+) = \lim_{h \rightarrow 0} \frac{\frac{\ln(1+2h)}{h} - 2}{h} = \lim_{h \rightarrow 0} \frac{\ln(1+2h) - 2h}{h^2} = \lim_{h \rightarrow 0} \frac{\left(2h - \frac{(2h)^2}{2} + \dots \right) - 2h}{h^2} = -2$$

$$\text{Also } f'(0^-) = \lim_{h \rightarrow 0} \frac{a \cot^{-1}\left(\frac{b-h}{4}\right) - 2}{-h} \left(\frac{0}{0}\right) = \lim_{h \rightarrow 0} \frac{\frac{-a}{1 + \left(\frac{b-h}{4}\right)^2} \times \frac{-1}{4}}{-1} = \frac{-4a}{b^2 + 16}$$

$$\text{As } f'(0^-) = f'(0^+), \text{ so } \frac{-4a}{b^2 + 16} = -2$$

$$\Rightarrow 2a = b^2 + 16$$

$$\therefore b^2 - 2a = -16 \quad \dots\dots\dots(2)$$

$$\text{Hence } (b^2 - 2a + c^6) = -16 + 64 = 48 \text{ (Using equation (1) and equation (2))}$$

DETERMINATION OF FUNCTION

34. If $|f(x_1) - f(x_2)| \leq (x_1 - x_2)^2$, for all $x_1, x_2 \in \mathbb{R}$. Find the equation of tangent to the curve $y = f(x)$ at the point (1, 2):

यदि $|f(x_1) - f(x_2)| \leq (x_1 - x_2)^2$, $x_1, x_2 \in \mathbb{R}$ तो वक्र $y = f(x)$ का (1, 2) पर स्पर्श रेखा का समीकरण बताओं :

Ans. $y - 2 = 0$

Sol. Put $x_1 = x + h$ & $x_2 = x$

$$|f(x+h) - f(x)| \leq h^2$$

$$\lim_{h \rightarrow 0} \left| \frac{f(x+h) - f(x)}{h} \right| \leq \lim_{h \rightarrow 0} h$$

$$|f'(x)| \leq 0$$

Possible only if $f'(x) = 0$

$$f(x) = c$$

at point (1, 2) $f(x) = 2$

$$y = 2$$

35. Let f be a function that is differentiable every where and that has the following properties:

(i) $f(x+h) = f(x) \cdot f(h)$ (ii) $f(x) > 0$ for all real x . (iii) $f'(0) = -1$

Use the definition of derivative to find $f'(x)$ in terms of $f(x)$:

यदि फलन f के सभी जगहों पर अवकलनीय है तथा निम्नलिखित गुण रखता है :

(i) $f(x+h) = f(x) \cdot f(h)$ (ii) $f(x) > 0 \forall x \in \mathbb{R}$ (iii) $f'(0) = -1$

तो अवकलनीयता की परिभाषा से $f'(x)$ का मान $f(x)$ के पदों में ज्ञात करो।

Ans. $f'(x) = -f(x)$

Sol. $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$$= \lim_{h \rightarrow 0} \frac{f(x)(f(h) - 1)}{h}$$

$$= f(x) \lim_{h \rightarrow 0} \frac{f(h) - 1}{h}$$

$$f(0+0) = (f(0))^2$$

$$\Rightarrow f(0) = 0 \quad \text{or} \quad f(0) = 1$$

$$\because f(x) > 0 \quad \forall x$$

$$\Rightarrow f(0) = 1$$

$$\Rightarrow f'(x) = f(x) \cdot f'(0) = -f(x)$$

$$\Rightarrow \ln(f(x)) = -x + c$$

$$\because f(0) = 1$$

$$\Rightarrow c = 0$$

$$\Rightarrow f(x) = e^{-x}$$

36. A derivable function $f: \mathbb{R}^+ \rightarrow \mathbb{R}$ satisfies the condition $f(x) - f(y) \geq \ln(x/y) + x - y$ for every $x, y \in \mathbb{R}^+$. If g

denotes the derivative of f then compute the value of the sum $\sum_{n=1}^{100} g\left(\frac{1}{n}\right)$:

एक अवकलनीय फलन $f: \mathbb{R}^+ \rightarrow \mathbb{R}$ समीकरण $f(x) - f(y) \geq \ln(x/y) + x - y$ जहाँ $x, y \in \mathbb{R}^+$ को सन्तुष्ट करता है यदि f के

अवकलन का मान g हो, तो $\sum_{n=1}^{100} g\left(\frac{1}{n}\right)$ का मान ज्ञात करो :

Ans. 5150

Sol.
$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\geq \lim_{h \rightarrow 0} \frac{\ell n\left(\frac{x+h}{x}\right) + x + h - x}{h}$$

$$\geq \lim_{h \rightarrow 0} \frac{\ell n\left(1 + \frac{h}{x}\right) + 1}{h} \geq \frac{1}{x} + 1 \quad \dots\dots\dots(i)$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x-h) - f(x)}{-h}$$

$$\leq \lim_{h \rightarrow 0} \frac{\ell n\left(\frac{x-h}{x}\right) + x - h - x}{-h}$$

$$\leq \lim_{h \rightarrow 0} \frac{\ell n\left(1 - \frac{h}{x}\right) + 1}{-h} \leq \frac{1}{x} + 1 \quad \dots\dots\dots(ii)$$

from (i) and (ii)

$$\Rightarrow f(x) = \frac{1}{x} + 1$$

$$\begin{aligned} \therefore \sum_{n=1}^{100} g\left(\frac{1}{n}\right) &= g\left(\frac{1}{1}\right) + g\left(\frac{1}{2}\right) + \dots + g\left(\frac{1}{100}\right) \\ &= (1 + 2 + 3 + \dots + 100) + 100 = 5150 \end{aligned}$$

Paragraph for Question nos. 37 to 38

Consider a function $f(x)$ in $[0, 2\pi]$ defined as :

$$f(x) = \begin{cases} [\sin x] + [\cos x] & ; \quad 0 \leq x \leq \pi \\ [\sin x] - [\cos x] & ; \quad \pi < x \leq 2\pi \end{cases}$$

where $[.]$ denotes greatest integer function then

माना $[0, 2\pi]$ में फलन $f(x)$ इस प्रकार परिभाषित है

$$f(x) = \begin{cases} [\sin x] + [\cos x] & ; \quad 0 \leq x \leq \pi \\ [\sin x] - [\cos x] & ; \quad \pi < x \leq 2\pi \end{cases}$$

जहां $[.]$ महत्तम पूर्णांक फलन को दर्शाता है

37. Number of points where $f(x)$ is non-derivable :

बिन्दुओं की संख्या जहां पर $f(x)$ अवकलनीय नहीं है, होगी :

(A) 2

(B) 3

(C) 4

(D) 5

Ans. C

Sol.

$$f(x) = \begin{cases} 1 & x = 0 \\ 0 & 0 < x < \frac{\pi}{2} \\ 1 & x = \frac{\pi}{2} \\ -1 & \frac{\pi}{2} < x \leq \pi \\ 0 & \pi < x < \frac{3\pi}{2} \\ -1 & x = \frac{3\pi}{2} \\ -1 & \frac{3\pi}{2} < x < 2\pi \\ -1 & x = 2\pi \end{cases}$$

$f(x)$ is not differentiable at $x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$

38. $\lim_{x \rightarrow \left(\frac{3\pi}{2}\right)^+} f(x)$ equals

$\lim_{x \rightarrow \left(\frac{3\pi}{2}\right)^+} f(x)$ का मान है -

- (A) 0 (B) 1 (C) -1 (D) 2

Ans. C

Sol. $\lim_{x \rightarrow \left(\frac{3\pi}{2}\right)^+} f(x) = -1$

Paragraph for Question nos. 39 to 41

Let $f(x) = \begin{cases} x[x] & 0 \leq x < 2 \\ (x-1)[x] & 2 \leq x \leq 3 \end{cases}$ where $[x]$ = greatest integer less than or equal to x , then

माना $f(x) = \begin{cases} x[x] & 0 \leq x < 2 \\ (x-1)[x] & 2 \leq x \leq 3 \end{cases}$ है जहाँ $[x]$ = महत्तम पूर्णांक फलन है तो

39. The number of values of x for $x \in [0, 3]$ where $f(x)$ is discontinuous is :

$x \in [0, 3]$ में बिन्दुओं की संख्या जहाँ पर $f(x)$ असतत है, होगी :

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. C

Sol.

40. The number of values of x for $x \in [0, 3]$ where $f(x)$ is non-differentiable is :

$x \in [0, 3]$ में बिन्दुओं की संख्या जहाँ पर $f(x)$ अवकलनीय नहीं है, होगी :

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. D

Sol.

41. The number of integers in the range of $y = f(x)$ is :

$y = f(x)$ के परिसर में पूर्णाकों की संख्या होगी :

- (A) 3 (B) 4 (C) 5 (D) 6

Ans. C

Sol.
$$f(x) = \begin{cases} 0 & 0 \leq x < 1 \\ x & 1 \leq x < 2 \\ 2(x-1) & 2 \leq x < 3 \\ 6 & x = 3 \end{cases}$$

$\therefore f(x)$ is discontinuous at $x = 1, 3$. $\rightarrow$ not differentiable at 1 & 3.

$$f'(x) = \begin{cases} 0 & 0 < x < 1 \\ 1 & 1 < x < 2 \\ 2 & 2 < x < 3 \end{cases}$$

$\therefore f(x)$ is not differentiable at $x = 1, 2, 3$.

$$R_f = [1, 4) \cup \{0\} \cup \{6\}$$

No. of integers = 5.

ONE OR MORE THAN ONE CORRECT TYPE

42. If $f(x) = \begin{cases} -x - \frac{\pi}{2}, & x \leq \frac{\pi}{2} \\ -\cos x, & -\frac{\pi}{2} < x \leq 0 \\ x - 1, & 0 < x \leq 1 \\ \ln x & x > 1 \end{cases}$ then :

[JEE 2011]

(A) $f(x)$ is continuous at $x = -\frac{\pi}{2}$

(B) $f(x)$ is not differentiable at $x = 0$

(C) $f(x)$ differentiable at $x = 1$

(D) $f(x)$ is differentiable at $x = -\frac{3}{2}$

यदि $f(x) = \begin{cases} -x - \frac{\pi}{2}, & x \leq \frac{\pi}{2} \\ -\cos x, & -\frac{\pi}{2} < x \leq 0 \\ x - 1, & 0 < x \leq 1 \\ \ln x & x > 1 \end{cases}$ तो :

[JEE 2011]

(A) $f(x)$, $x = -\frac{\pi}{2}$ पर सतत् होगा

(B) $f(x)$, $x = 0$ पर अवकलनीय नहीं होगा

- (C) $f(x)$, $x = 1$ पर अवकलनीय होगा (D) $f(x)$, $x = -\frac{3}{2}$ पर अवकलनीय होगा

Ans. ABCD

Sol. $1\left(-\frac{\pi}{2}\right) = 0$, $1\left(-\frac{\pi}{2}\right) = 0$

$$f'(x) = \begin{cases} -1 & x \leq \frac{\pi}{2} \\ \sin x & -\frac{\pi}{2} < x \leq 0 \\ 1 & 0 < x \leq 1 \\ \frac{1}{x} & x > 1 \end{cases}$$

$f'(0^-) = 0$, $f'(0^+) = 1 \therefore$ not differentiable at $x = 0$

$f'(1^-) = 1$, $f'(1^+) = 1 \therefore$ differentiable at $x = 1$

as $-\frac{3}{2} \in \left(-\frac{\pi}{2}, 0\right)$

$f'(x) = \sin x$ which is differentiable at $x = -\frac{3}{2}$

43. Which one of the following statements is NOT CORRECT :

- (A) The derivative of a differentiable periodic function is a periodic function with the same period
 (B) If $f(x)$ and $g(x)$ both are defined on the entire number line and are aperiodic then the function $F(x) = f(x) \cdot g(x)$ can not be periodic
 (C) Derivative of an even differentiable function is an odd function and derivative of an odd differentiable function is an even function
 (D) Every function $f(x)$ can be represented as the sum of an even and an odd function

निम्न में से कौनसा कथन सही नहीं है :

- (A) किसी अवकलनीय आवर्ती फलन का अवकलज उसी आवर्त में अवकलनीय होता है।
 (B) यदि $f(x)$ तथा $g(x)$ सम्पूर्ण संख्या रेखा पर परिभाषित है तथा आवर्ती नहीं है तो $F(x) = f(x) \cdot g(x)$ आवर्ती नहीं हो सकते
 (C) सम अवकलनीय फलन का अवकलज विषम होता है तथा विषम अवकलनीय फलन का अवकलज सम होता है।
 (D) प्रत्येक फलन $f(x)$ एक सम तथा विषम फलन के योग के रूप में व्यक्त किया जा सकता है।

Ans. BD

Sol. (A) $f(x+T) = f(x) \forall x \in \mathbb{R}$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\Rightarrow f'(x+T) = \lim_{h \rightarrow 0} \frac{f(x+h+T) - f(x+T)}{h}$$

$$= f'(x)$$

$$(B) \left. \begin{array}{l} f(x) = e^x \\ g(x) = e^{-x} \end{array} \right\} \rightarrow \text{aperiodic}$$

$$\Rightarrow f(x) \cdot g(x) = 1 \rightarrow \text{Periodic}$$

$$(C) f(-x) = f(x)$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(-x) = \lim_{h \rightarrow 0} \frac{f(-x+h) - f(-x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x-h) - f(x)}{h}$$

$$= -f'(x)$$

(D) Domain should be symmetric about origin.

Take $f(x) = \log_e x$.

44. If $f(x) = \begin{cases} \frac{x \cdot \ln(\cos x)}{\ln(1+x^2)} & x \neq 0 \\ 0 & x = 0 \end{cases}$ then :

(A) f is continuous at $x = 0$

(B) f is continuous at $x = 0$ but not differentiable at $x = 0$

(C) f is differentiable at $x = 0$

(D) f is not continuous at $x = 0$

$$\text{यदि } f(x) = \begin{cases} \frac{x \cdot \ln(\cos x)}{\ln(1+x^2)} & x \neq 0 \\ 0 & x = 0 \end{cases} \text{ तो :}$$

(A) f , $x = 0$ पर संतत है

(B) f , $x = 0$ पर संतत है परन्तु अवकलनीय नहीं है :

(C) f , $x = 0$ अवकलनीय है

(D) f , $x = 0$ संतत नहीं है

Ans. AC

Sol. $f'(0^+) = \lim_{h \rightarrow 0} \frac{h \ln(\cosh)}{h \ln(1+h^2)} = \lim_{h \rightarrow 0} \frac{\ln(\cosh)^{1/h^2}}{\ln(1+h^2)} = \lim_{h \rightarrow 0} \frac{1}{h^2} (\cosh - 1) = -\frac{1}{2}; \quad \text{||ly } f'(0^-) = -\frac{1}{2}$

hence f is continuous and derivable at $x = 0$

45. Let $f(x)$ be defined as follows: $f(x) = \begin{cases} x^6, & x^2 > 1 \\ x^3, & x^2 \leq 1 \end{cases}$. Then $f(x)$ is:

- (A) continuous everywhere (B) differentiable everywhere
(C) discontinuous at $x = -1$ (D) not differentiable at $x = 1$

यदि $f(x) = \begin{cases} x^6, & x^2 > 1 \\ x^3, & x^2 \leq 1 \end{cases}$ तो $f(x)$ होगा :

- (A) सभी बिन्दुओं पर संतत है (B) सभी बिन्दुओं पर अवकलनीय है
(C) $x = -1$ पर असंतत है (D) $x = 1$ पर अवकलनीय नहीं है

Ans. CD

Sol. $f(x) = \begin{cases} x^6 & x^2 > 1 \\ x^3 & x^2 \leq 1 \end{cases}$

$$f(x) = \begin{cases} x^6 & x < -1 \\ x^3 & -1 \leq x \leq 1 \\ x^6 & x > 1 \end{cases}$$

$\therefore f(x)$ is discontinuous at $x = 1$.

at $x = 1$

$$f'(x) = \begin{cases} 3x^2 & x < 1 \\ 6x^5 & x > 1 \end{cases}$$

$$f'(1^+) = 6$$

$$f'(1^-) = 3$$

$\Rightarrow f(x)$ is not differentiable at $x = 1$.

46. Which of the following functions f has/have a removable discontinuity at the indicated point :

(A) $f(x) = \frac{x^2 - 2x - 8}{x + 2}$ at $x = -2$

(B) $f(x) = \frac{x - 7}{|x - 7|}$ at $x = 7$

(C) $f(x) = \frac{x^3 + 64}{x + 4}$ at $x = -4$

(D) $f(x) = \frac{3 - \sqrt{x}}{9 - x}$ at $x = 9$

दिये गये बिन्दु पर निम्नलिखित फलन में कौन सा फलन ऐसा जिसमें विस्थापनीय असतत्ता है :

(A) $f(x) = \frac{x^2 - 2x - 8}{x + 2}$, $x = -2$ पर

(B) $f(x) = \frac{x - 7}{|x - 7|}$, $x = 7$ पर

(C) $f(x) = \frac{x^3 + 64}{x + 4}$, $x = -4$ पर

(D) $f(x) = \frac{3 - \sqrt{x}}{9 - x}$, $x = 9$ पर

Ans. ACD

Sol. (A) $f(x) = \frac{x^2 - 2x - 8}{x + 2} = \frac{(x + 2)(x - 4)}{(x + 2)}$

LHL = -6 = RHL

$f(-2)$ = Not defined $\rightarrow$ Removable discontinuity

(B) $f(x) = \frac{x - 7}{|x - 7|}$

at $x = 7$

LHL = -1 $\rightarrow$ Non removable

RHL = 1

(C) $f(x) = \frac{x^3 + 64}{x + 4} \quad x = -4$

$\left. \begin{array}{l} \text{LHL} = 16 = \text{RHL} \\ f(-4) = \text{not defined} \end{array} \right\} \rightarrow \text{Removable discontinuity}$

(D) $f(x) = \frac{3 - \sqrt{x}}{9 - x}$

at $x = 9$

$\left. \begin{array}{l} \text{LHL} = \frac{1}{6} = \text{RHL} \\ f(9) = \text{not defined} \end{array} \right\} \rightarrow \text{Removable discontinuity}$

47. In which of the following cases the given equations has atleast one root in the indicated interval :

(A) $x - \cos x = 0$ in $(0, \pi/2)$

(B) $x + \sin x = 1$ in $(0, \pi/6)$

(C) $\frac{a}{x-1} + \frac{b}{x-3} = 0, \quad a, b > 0$ in $(1, 3)$

(D) $f(x) - g(x) = 0$ in (a, b) where f and g are continuous on $[a, b]$ and $f(a) > g(a)$ and $f(b) < g(b)$

निम्न में से कौन से फलन का दिये गये संगत अन्तराल में कम से कम एक मूल होगा :

(A) $x - \cos x = 0, (0, \pi/2)$ में

(B) $x + \sin x = 1, (0, \pi/6)$ में

(C) $\frac{a}{x-1} + \frac{b}{x-3} = 0, \quad a, b > 0, (1, 3)$ में

(D) $f(x) - g(x) = 0, (a, b)$ में जबकि f तथा g अन्तराल $[a, b]$ में सतत् हो तथा $f(a) > g(a)$ और $f(b) < g(b)$.

Ans. ABCD

Sol. $f(x) = x - \cos x$

$x \in \left(0, \frac{\pi}{2}\right) \rightarrow f(x)$ is continuous & differentiable

$$f(0) = -1$$

$$f\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$$

$$\Rightarrow f(0) \cdot f\left(\frac{\pi}{2}\right) < 0$$

$$\Rightarrow \exists c \in \left(0, \frac{\pi}{2}\right) \text{ s.t. } f(c) = 0$$

$$\Rightarrow c - \cos c = 0$$

(B) $f(x) = x + \sin x - 1$

$x \in \left(0, \frac{\pi}{6}\right) \rightarrow f(x)$ is continuous & differentiable

$$f(0) = -1$$

$$f\left(\frac{\pi}{6}\right) = \frac{\pi}{6} - \frac{1}{2} = \frac{\pi-3}{6}$$

$$\Rightarrow f(0) \cdot f\left(\frac{\pi}{6}\right) < 0$$

$$\Rightarrow \exists c \in \left(0, \frac{\pi}{6}\right) \text{ s.t. } f(c) = 0$$

(C) $f(x) = \frac{a}{x-1} + \frac{b}{x-3}, a, b > 0$

in $(1, 3) \rightarrow f(x)$ is continuous & differentiable

$$f(1^+) \rightarrow \infty$$

$$f(3^-) \rightarrow \infty$$

$$\Rightarrow \exists c \in (1, 3) \text{ s.t. } \frac{a}{c-1} + \frac{b}{c-3} = 0$$

(D) $f(x)$ and $g(x)$ are continuous on $[a, b]$.

Let $h(x) = f(x) - g(x) \rightarrow$ continuous on $[a, b]$.

$$h(a) > 0 \text{ \& } h(b) < 0$$

$$\Rightarrow \exists c \in (a, b) \text{ s.t. } h(c) = 0$$

48. Indicate all correct alternatives if, $f(x) = \frac{x}{2} - 1$, then on the interval $[0, \pi]$:

(A) $\tan(f(x))$ & $\frac{1}{f(x)}$ are both continuous (B) $\tan(f(x))$ & $\frac{1}{f(x)}$ are both discontinuous

(C) $\tan(f(x))$ & $f^{-1}(x)$ are both continuous (D) $\tan(f(x))$ is continuous but $\frac{1}{f(x)}$ is not

यदि $f(x) = \frac{x}{2} - 1$, $[0, \pi]$ तो सही विकल्प को चुनो :

(A) $\tan(f(x))$ तथा $\frac{1}{f(x)}$ दोनों सतत् है। (B) $\tan(f(x))$ तथा $\frac{1}{f(x)}$ दोनों असतत् है।

(C) $\tan(f(x))$ तथा $f^{-1}(x)$ दोनों सतत् है। (D) $\tan(f(x))$ सतत् है लेकिन $\frac{1}{f(x)}$ सतत् नहीं है।

Ans. CD

Sol. (i) $\tan f(x) = \tan\left(\frac{x}{2} - 1\right)$ $x \in [0, \pi]$

$$0 \leq x \leq \pi \Rightarrow -1 \leq \frac{x}{2} - 1 \leq \frac{\pi}{2} - 1$$

By graph we say $\tan(f(x))$ is continuous in $[0, \pi]$

(ii) $\frac{1}{f(x)} = \frac{2}{x-2}$ is not defined at $x = 2 \in [0, \pi]$

$$(iii) y = \frac{x-2}{2}$$

$\Rightarrow f^{-1}(x) = 2x + 2$ is continuous in $\mathbb{R}$.

49. Which of the following function(s) not defined at $x = 0$ has/have non-removable discontinuity at the point $x = 0$:

निम्न में से कौनसे फलन $x = 0$ पर परिभाषित नहीं है तथा $x = 0$ पर अविस्थापनीय असतत्ता है :

(A) $f(x) = \frac{1}{1 + 2^{\frac{1}{x}}}$ (B) $f(x) = \arctan \frac{1}{x}$ (C) $f(x) = \frac{e^{\frac{1}{x}} - 1}{e^{\frac{1}{x}} + 1}$ (D) $f(x) = \frac{1}{\ln|x|}$

Ans. ABC

Sol. (A) $f(x) = \frac{1}{1 + 2^{\frac{1}{x}}}$ $\rightarrow$ not defined at $x = 0$

$$\text{LHL} = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} \frac{1}{1 + 2^{-\frac{1}{h}}} = 1$$

$$\text{RHL} = \lim_{h \rightarrow 0} \frac{1}{1 + 2^{\frac{1}{h}}} = 0$$

$\Rightarrow f(x)$ has non removable discontinuous at $x = 0$.

(B) $f(x) = \tan^{-1}\left(\frac{1}{x}\right) \rightarrow$ not defined at $x = 0$.

$$\left. \begin{aligned} \text{LHL} &= -\lim_{h \rightarrow 0} \tan^{-1}\left(\frac{1}{h}\right) = -\frac{\pi}{2} \\ \text{RHL} &= \lim_{h \rightarrow 0} \tan^{-1}\left(\frac{1}{h}\right) = \frac{\pi}{2} \end{aligned} \right\} \rightarrow \text{non-removable discontinuity}$$

(C) $f(x) = \frac{e^{\frac{1}{x}} - 1}{\frac{1}{e^x + 1}} \rightarrow$ not defined at $x = 0$

$$\left. \begin{aligned} \text{LHL} &= \lim_{h \rightarrow 0} \frac{e^{\frac{1}{h}} - 1}{\frac{1}{e^h + 1}} = -1 \\ \text{RHL} &= \lim_{h \rightarrow 0} \frac{e^{-\frac{1}{h}} - 1}{\frac{1}{e^{-h} + 1}} = 1 \end{aligned} \right\} \rightarrow \text{non removable discontinuity}$$

(D) $f(x) = \frac{1}{\ln|x|} \rightarrow$ not defined at $x = 0$

$$\text{LHL} = \lim_{h \rightarrow 0} \frac{1}{\ln|h|} = \text{RHL} = 0 \rightarrow \text{removable discontinuity}$$

50. The function, $f(x) = [x] - |[x]|$ where $[x]$ denotes greatest integer function :

- (A) is continuous for all positive integers
- (B) is discontinuous for all non positive integers
- (C) has finite number of elements in its range
- (D) is such that its graph does not lie above the x -axis

फलन $f(x) = [x] - |[x]|$ है जहां $[x]$ महत्तम पूर्णांक फलन को दर्शाता है :

- (A) सभी धनात्मक पूर्णांक मानों के लिए सतत है
- (B) सभी अधनात्मक पूर्णांक मानों के लिए असतत है
- (C) इसके परिसर में परिमित संख्या में अवयव है
- (D) इस प्रकार है कि इसका आरेख x -अक्ष के ऊपर विद्यमान नहीं है

Ans. ABCD

Sol. $f(x) = [x] - |[x]|$

Let $x = \alpha \in I^+$

$$\lim_{h \rightarrow 0} f(\alpha + h) = [\alpha + h] - \lceil \alpha + h \rceil$$

$$= \alpha - |\alpha| = 0$$

$$\lim_{h \rightarrow 0} f(\alpha - h) = \lceil \alpha - h \rceil - [\alpha - h]$$

$$= (\alpha - 1) - (\alpha - 1)$$

$$f(\alpha) = 0$$

$\therefore f(x)$ is continuous for all positive integer

Let $x = \alpha \in I^-$

$$f(\alpha) = \lceil \alpha \rceil - \lfloor \alpha \rfloor$$

$$= |\alpha| - \alpha = 0$$

$$\lim_{h \rightarrow 0} f(\alpha - h) = \lim_{h \rightarrow 0} \lceil \alpha - h \rceil - \lfloor \alpha - h \rfloor$$

$$= \lim_{h \rightarrow 0} [h - \alpha] - |\alpha - 1|$$

$$= (-\alpha) - (1 - \alpha)$$

$$= -1$$

$$\lim_{h \rightarrow 0} f(\alpha + h) = \lim_{h \rightarrow 0} \lceil \alpha + h \rceil - \lfloor \alpha + h \rfloor$$

$$= \lim_{h \rightarrow 0} [-h - \alpha] - |\alpha|$$

$$= (-\alpha - 1) + \alpha = -1$$

$\therefore f(x)$ is discontinuous $\forall \alpha \in I$

Now $f(x) = \lceil x \rceil - \lfloor x \rfloor$

$$= \begin{cases} \lceil x \rceil - \lfloor x \rfloor & x \geq 0 \\ \lceil -x \rceil - \lfloor x \rfloor & x < 0 \end{cases}$$

$$= \begin{cases} 0 & x \geq 0 \\ \lceil -x \rceil + \lfloor x \rfloor & x < 0 \end{cases}$$

$$= \begin{cases} 0 & x \geq 0 \\ -1 & -1 < x < 0 \\ -1 & -2 < x < -1 \end{cases}$$

$$\Rightarrow R_f = \{0, -1\}$$

Graph of $f(x)$ does not lie above the x-axis.

51. Given that the derivative $f'(a)$ exists. Indicate which of the following statement(s) is/are always True :

दिया है कि $f'(a)$ का अस्तित्व है, तो निम्न में से कौन सा कथन हमेशा सत्य होगा :

$$(A) f'(a) = \lim_{h \rightarrow a} \frac{f(h) - f(a)}{h - a}$$

$$(B) f'(a) = \lim_{h \rightarrow 0} \frac{f(a) - f(a - h)}{h}$$

$$(C) f'(a) = \lim_{t \rightarrow 0} \frac{f(a + 2t) - f(a)}{t}$$

$$(D) f'(a) = \lim_{t \rightarrow 0} \frac{f(a + 2t) - f(a + t)}{2t}$$

Ans. AB

Sol. (C) is false and is True only if $f'(a) = 0$ limit is $2f'(a)$. In (D) same logic limit is $\frac{1}{2}f'(a)$

52. The function $f(x) = \sqrt{1 - \sqrt{1 - x^2}}$:

(A) has its domain $-1 \leq x \leq 1$.

(B) has finite one sided derivatives at the point $x = 0$.

(C) is continuous and differentiable at $x = 0$.

(D) is continuous but not differentiable at $x = 0$

दिया है $f(x) = \sqrt{1 - \sqrt{1 - x^2}}$ तो फलन $f(x)$:

(A) का प्रान्त $-1 \leq x \leq 1$ होगा ।

(B) का $x = 0$ पर परिमित तथा एक तरफा अवकलन होगा ।

(C) $x = 0$ पर सतत् तथा अवकलनीय होगा ।

(D) $x = 0$ पर फलन सतत् तो होगा लेकिन अवकलनीय नहीं होगा ।

Ans. ABD

Sol. $f'(0^+) = \frac{1}{\sqrt{2}}; f'(0^-) = -\frac{1}{\sqrt{2}}; f(x) = \frac{\sqrt{x^2}}{\sqrt{1 + \sqrt{1 - x^2}}} = \frac{|x|}{\sqrt{1 + \sqrt{1 - x^2}}}$]

53. Let $f: (0, \infty) \rightarrow [1, \infty)$ be defined as $f(x) = \begin{cases} \frac{1}{x}, & 0 < x < 1 \\ \frac{x}{[x]}, & 1 \leq x < \infty \end{cases}$:

where $[x]$ denotes largest integer less than or equal to x . Then which of the following statements is (are) correct ?

(A) $f(x)$ is neither injective nor surjective.

(B) $f(x)$ is continuous but non-derivable at $x = 1$.

(C) $f(x)$ has non-removable discontinuity of finite type at $x = 2$ with jump equals $\frac{1}{2}$.

(D) $f'(e) = \frac{1}{2}$

माना $f: (0, \infty) \rightarrow [1, \infty)$ परिभाषित है $f(x) = \begin{cases} \frac{1}{x}, & 0 < x < 1 \\ \frac{x}{[x]}, & 1 \leq x < \infty \end{cases}$:

जहाँ $[x]$ महत्तम पूर्णांक फलन है तो निम्न में से कौनसे कथन सत्य है ?

(A) $f(x)$ ना तो एकेकी है ना ही आच्छादक

(B) $f(x)$, $x = 1$ पर सतत् है लेकिन अवकलनीय नहीं

(C) $f(x)$, $x = 2$ पर एक निश्चित प्रकार के अविस्थापनीय असतत्ता रखता है जहाँ $\frac{1}{2}$ का उछाल है

(D) $f'(e) = \frac{1}{2}$.

Ans. BD

Sol. (A) Clearly $f\left(\frac{2}{3}\right) = \frac{3}{2}$ and $f\left(\frac{3}{2}\right) = \frac{3}{2}$.

So, $f(x)$ is many one function.

Also range of $f(x) = [1, \infty)$, so

$f(x)$ is onto also. $\Rightarrow$ (A) is incorrect]

(B) $f(1^-) = 1 = f(1^+) = f(1)$

But $f'(1^-) = -1$ and $f'(1^+) = 1$

So, $f(x)$ is continuous but non-derviable at $x = 1 \Rightarrow$ (B) is correct.

(C) $f(2^+) = 1$, $f(2^-) = 2$

$\therefore \lim_{x \rightarrow 2} f(x)$ does not exist.

Also jump of discontinuity $= |2 - 1| = 1 \Rightarrow$ (C) is correct

(D) Clearly $f'(e) = \frac{1}{2}$ (verify)]

54. $f(x) = |x[x]|$ in $-1 \leq x \leq 2$, where $[x]$ is greatest integer $\leq x$ then $f(x)$ is :

(A) continuous at $x = 0$

(B) discontinuous $x = 0$

(C) not differentiable at $x = 2$

(D) differentiable at $x = 2$

यदि $f(x) = |x[x]|$, $-1 \leq x \leq 2$, जहाँ $[x]$ महत्तम पूर्णांक फलन हो, तो $f(x)$:

(A) $x = 0$ पर सतत् होगा।

(B) $x = 0$ पर असतत् होगा

(C) $x = 2$ पर अवकलनीय नहीं होगा

(D) $x = 2$ पर अवकलनीय होगा

Ans. AC

Sol. $f(x) = |x[x]|$ $-1 \leq x \leq 2$

$$= \begin{cases} |-x| & -1 \leq x < 0 \\ 0 & 0 \leq x < 1 \\ |x| & 1 \leq x < 2 \\ 4 & x = 2 \end{cases}$$

$$= \begin{cases} (-x) & -1 \leq x < 0 \\ 0 & 0 \leq x < 1 \\ x & 1 \leq x < 2 \\ 4 & x = 2 \end{cases}$$

$f(x)$ is continuous at $x = 0$.

$\therefore f(x)$ is not differentiable at $x = 2$ as $f(x)$ is not continuous at $x = 2$.

55. $f(x) = 1 + x \cdot [\cos x]$ in $0 < x \leq \pi/2$, where $[]$ denotes greatest integer function then :
- (A) It is continuous in $0 < x < \pi/2$ (B) It is differentiable in $0 < x < \pi/2$
 (C) Its maximum value is 2 (D) It is not differentiable in $0 < x < \pi/2$
- यदि $f(x) = 1 + x \cdot [\cos x]$ $0 < x \leq \pi/2$, जहाँ $[]$ महत्तम पूर्णांक फलन हो, तो :
- (A) फलन $0 < x < \pi/2$ में सतत् है। (B) फलन $0 < x < \pi/2$ में अवकलनीय है।
 (C) फलन का अधिकतम मान 2 होगा (D) फलन $0 < x < \pi/2$ में अवकलनीय नहीं होगा।

Ans. AB

Sol. $f(x) = 1 + x \cdot [\cos x]$; $0 < x \leq \frac{\pi}{2}$

$$\Rightarrow f(x) = 1 \rightarrow \text{continuous \& differentiable in } \left(0, \frac{\pi}{2}\right)$$

56. Select the correct statements :

- (A) The function f defined by $f(x) = \begin{cases} 2x^2 + 3 & \text{for } x \leq 1 \\ 3x + 2 & \text{for } x > 1 \end{cases}$ is neither differentiable nor continuous at $x = 1$.
 (B) The function $f(x) = x^2 |x|$ is twice differentiable at $x = 0$.
 (C) If f is continuous at $x = 5$ and $f(5) = 2$ then $\lim_{x \rightarrow 2} f(4x^2 - 11)$ exists.
 (D) If $\lim_{x \rightarrow a} (f(x) + g(x)) = 2$ and $\lim_{x \rightarrow a} (f(x) - g(x)) = 1$ then $\lim_{x \rightarrow a} f(x) \cdot g(x)$ need not exist

सही कथन का चुनाव कीजिये :

- (A) f द्वारा परिभाषित फलन $f(x) = \begin{cases} 2x^2 + 3 & \text{for } x \leq 1 \\ 3x + 2 & \text{for } x > 1 \end{cases}$, $x = 1$ पर ना तो अवकलनीय और ना ही सतत्
 (B) फलन $f(x) = x^2 |x|$, $x = 0$ पर दो बार अवकलनीय है

(C) यदि $f, x = 5$ पर सतत है तथा $f(5) = 2$ है तो $\lim_{x \rightarrow 2} f(4x^2 - 11)$ विद्यमान है

(D) यदि $\lim_{x \rightarrow a} (f(x) + g(x)) = 2$ तथा $\lim_{x \rightarrow a} (f(x) - g(x)) = 1$ है तो $\lim_{x \rightarrow a} f(x) \cdot g(x)$ आवश्यक नहीं है कि विद्यमान हो

Ans. BC

Sol. (A) $f(x) = \begin{cases} 2x^2 + 3 & x \leq 1 \\ 3x + 2 & x > 1 \end{cases}$

at $x = 1$; $f(x)$ is continuous & not differentiable

(B) $f(x) = \begin{cases} -x^3 & x < 0 \\ x^3 & x \geq 0 \end{cases}$

$f'(x) = \begin{cases} -3x^2 & x < 0 \\ 3x^2 & x \geq 0 \end{cases}$

$f''(x) = \begin{cases} -6x & x < 0 \\ 6x & x > 0 \end{cases}$

$\therefore f(x)$ is twice differentiable at $x = 0$.

(C) $\lim_{x \rightarrow 2} f(4x^2 - 11) = f(5) = 2$

(D) $\lim_{x \rightarrow a} [(f(x) + g(x)) + f(x) - g(x)] = 3$

$\lim_{x \rightarrow a} f(x) = \frac{3}{2} \quad \& \quad \lim_{x \rightarrow b} g(x) = \frac{1}{2}$

$\Rightarrow \lim_{x \rightarrow a} f(x) \cdot g(x) = \frac{3}{4}$

57. Let $f(x) = |2x - 9| + |2x| + |2x + 9|$. Which of the following are true :

(A) $f(x)$ is not differentiable at $x = \frac{9}{2}$

(B) $f(x)$ is not differentiable at $x = \frac{-9}{2}$

(C) $f(x)$ is not differentiable at $x = 0$

(D) $f(x)$ is differentiable at $x = \frac{-9}{2}, 0, \frac{9}{2}$

माना $f(x) = |2x - 9| + |2x| + |2x + 9|$ है तो निम्न में से कौनसा सत्य है :

(A) $x = \frac{9}{2}$ पर $f(x)$ अवकलनीय नहीं है

(B) $x = \frac{-9}{2}$ पर $f(x)$ अवकलनीय नहीं है

(C) $x = 0$ पर $f(x)$ अवकलनीय नहीं है

(D) $x = \frac{-9}{2}, 0, \frac{9}{2}$ पर $f(x)$ अवकलनीय है

Ans. ABC

Sol. $f(x) = \underbrace{|2x-9|}_{x=\frac{9}{2}} + \underbrace{|2x|}_{x=0} + \underbrace{|2x+9|}_{x=-\frac{9}{2}}$

$$= \begin{cases} -2x+9-2x-2x-9 & x < -\frac{9}{2} \\ -2x+9-2x+2x+9 & -\frac{9}{2} \leq x < 0 \\ -2x+9+2x+2x+9 & 0 \leq x < \frac{9}{2} \\ 6x-9+9 & x \geq \frac{9}{2} \end{cases}$$

$$= \begin{cases} -6x & x < -\frac{9}{2} \\ -2x+18 & -\frac{9}{2} \leq x < 0 \\ 2x+18 & 0 \leq x < \frac{9}{2} \\ 6x & x \geq \frac{9}{2} \end{cases}$$

$\therefore f(x)$ is continuous $\forall x \in \mathbb{R}$

$$f'(x) = \begin{cases} -6 & x < -\frac{9}{2} \\ -2 & -\frac{9}{2} < x < 0 \\ 2 & 0 < x < \frac{9}{2} \\ 6 & x > \frac{9}{2} \end{cases}$$

$\therefore f(x)$ is not differentiable at $-\frac{9}{2}, 0, \frac{9}{2}$

MATCH THE COLUMN

58. **Column-I** **Column-II**
- (A) If $f(x)$ is derivable at $x = 3$ & $f'(3) = 2$, then $\lim_{h \rightarrow 0} \frac{f(3+h^2) - f(3-h^2)}{2h^2}$ equals (P) 0
- (B) Let $f(x)$ be a function satisfying the condition $f(-x) = f(x)$ for all real x . If $f'(0)$ exists, then its value is equal to (Q) 1
- (C) For the function $f(x) = \begin{cases} \frac{x}{1+e^{1/x}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$, the derivative from the left, $f'(0^-)$ equals (R) 2
- (D) The number of points at which the function $f(x) = \max. \{a-x, a+x, b\}$, $-\infty < x < \infty$, $0 < a < b$ cannot be differentiable is (S) 3

स्तम्भ -I

(A) यदि $f(x)$, $x = 3$ पर अवकलनीय है तथा $f'(3) = 2$,

तो $\lim_{h \rightarrow 0} \frac{f(3+h^2) - f(3-h^2)}{2h^2}$ का मान होगा।

(B) यदि x के सभी वास्तविक मानों के लिये $f(x) = f(-x)$ है तो $f'(0)$ का मान ज्ञात करें।

(C) फलन $f(x) = \begin{cases} \frac{x}{1+e^{1/x}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$, में $f'(0^-)$ का मान होगा

(D) फलन $f(x) = \max \{a-x, a+x, b\}$, $-\infty < x < \infty$, $0 < a < b$ कितने बिन्दुओं पर अवकलनीय नहीं होगा।

स्तम्भ-II

(P) 0

(Q) 1

(R) 2

(S) 3

Ans. (A) R, (B) P, (C) Q, (D) R

Sol. (A) $\lim_{h \rightarrow 0} \frac{f(3+h)^2 - f(3-h)^2}{2h^2} = f'(3) = 2$

(B) $f'(0) = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h}$

$= -\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$

$= -f'(0)$

$\Rightarrow f'(0) = 0$

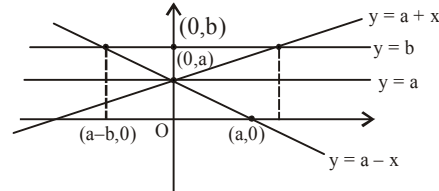
(C) $f'(0^-) = \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h}$

$= \lim_{h \rightarrow 0} \frac{-h}{\frac{1}{1+e^{-1/h}}(-h)} = 1$

$$(D) f(x) = \max\{a-x, a+x, b\} \quad 0 < a < b$$

$$= \begin{cases} a-x & x < a-b \\ b & a-b \leq x < b-a \\ a+x & x \geq b-a \end{cases}$$

$$f'(x) = \begin{cases} -1 & x < a-b \\ 0 & a-b < x < b-a \\ 1 & x > b-a \end{cases}$$



$\therefore f(x)$ is not differentiable at $x = a-b, b-a$.

59. **Column-I** contains 4 functions and **column-II** contains comments w.r.t their continuity and differentiability at $x = 0$. Note that **column-I** may have more than one matching options in **column-II**.

Column-I

(A) $f(x) = [x] + |1-x|$ where $[]$ denotes the greatest integer function

(B) $g(x) = |x-2| + |x|$

(C) $h(x) = [\tan^2 x]$ where $[]$ denotes the greatest integer function

(D) $l(x) = \begin{cases} \frac{x(3e^{1/x} + 4)}{(2 - e^{1/x})} & x \neq 0 \\ 0 & x = 0 \end{cases}$

Column-II

(P) continuous

(Q) differentiable

(R) discontinuous

(S) non derivable

स्तम्भ-I में चार फलन दिये हैं तथा **स्तम्भ-II** में फलन की $x = 0$ पर सतत्ता एवं अवकलनीय पर टिप्पणी की गयी है। ध्यान देने की बात है कि **स्तम्भ-I** के किसी भी विकल्प के लिये **स्तम्भ-II** में एक या एक से अधिक मिलान सम्भव है।

स्तम्भ-I

(A) $f(x) = [x] + |1-x|$

जहाँ $[]$ महत्तम पूर्णांक फलन है।

(B) $g(x) = |x-2| + |x|$

(C) $h(x) = [\tan^2 x]$

जहाँ $[]$ महत्तम पूर्णांक फलन है।

(D) $l(x) = \begin{cases} \frac{x(3e^{1/x} + 4)}{(2 - e^{1/x})} & x \neq 0 \\ 0 & x = 0 \end{cases}$

स्तम्भ-II

(P) सतत्

(Q) अवकलनीय

(R) असतत्

(S) अवकलनीय नहीं है।

Ans. (A) R, S; (B) P, S; (C) P, Q; (D) P, S

Sol. $f(x) = [x] + |1 - x|$

$$(A) = \begin{cases} [x] + 1 - x & x < 0 \\ 1 - x & 0 \leq x < 1 \\ 1 & x = 1 \\ x - 1 + [x] & x > 1 \end{cases}$$

$$= \begin{cases} [x] + 1 - x & x < -1 \\ -x & -1 \leq x < 0 \\ 1 - x & 0 \leq x < 1 \\ 1 & x = 1 \\ x - 1 + [x] & x > 1 \end{cases}$$

$\therefore f(x)$ is discontinuous at $x = 0$.

is not differentiable at $x = 0$.

(B) $g(x) = |x - 2| + |x|$

$$= \begin{cases} -2x^2 + 2 & x < 0 \\ 2 & x = 0 \\ 2 & 0 < x \leq 2 \\ 2x - 2 & x > 2 \end{cases}$$

$\therefore g(x)$ is continuous at $x = 0$.

$$g'(x) = \begin{cases} -2 & x < 0 \\ 2 & x > 0 \end{cases}$$

$\therefore g(x)$ is not differentiable at $x = 0$.

(C) $h(x) = [\tan^2 x]$

$$= \begin{cases} 0 & -\frac{\pi}{4} < x < 0 \\ 0 & x = 0 \\ 0 & 0 < x < \frac{\pi}{4} \end{cases}$$

$\therefore h(x)$ is differentiable and continuous at $x = 0$.

$$(D) \lim_{x \rightarrow 0^+} \frac{x \left(3e^{\frac{1}{x}} + 4 \right)}{2 - e^{\frac{1}{x}}} = 0 \text{ \& } f(0) = 0$$

$$\lim_{x \rightarrow 0^-} \frac{x \left(\frac{1}{3e^x + 4} \right)}{2 - e^x} = 0$$

$\therefore f(x)$ is continuous at $x = 0$.

$$\text{LHD} = -2$$

$$\text{RHD} = -3$$

$\therefore f(x)$ is not differentiable at $x = 0$.

60. Match the following column :

Column-I

(A) $f(x) = \begin{cases} x+1 & \text{if } x < 0 \\ \cos x & \text{if } x \geq 0 \end{cases}$ at $x = 0$ is

(B) For every $x \in \mathbb{R}$ the function $g(x) = \frac{\sin(\pi[x - \pi])}{1 + [x]^2}$

where $[x]$ denotes the greatest integer function is

(C) $h(x) = \sqrt{\{x\}^2}$ where $\{x\}$ denotes fractional part function $\forall x \in \mathbb{I}$, is

(D) $k(x) = \begin{cases} x^{\frac{1}{\ln x}} & \text{if } x \neq 1 \\ e & \text{if } x = 1 \end{cases}$ at $x = 1$ is

निम्न कॉलम को सुमेलित कीजिए :

Column-I

(A) $f(x) = \begin{cases} x+1 & \text{if } x < 0 \\ \cos x & \text{if } x \geq 0 \end{cases}$, $x = 0$ पर

(B) x के प्रत्येक मान के लिये $g(x) = \frac{\sin(\pi[x - \pi])}{1 + [x]^2}$, $x \in \mathbb{R}$

जहाँ $[x]$ महत्तम पूर्णांक फलन है।

(C) $h(x) = \sqrt{\{x\}^2}$ जहाँ $\{x\}$ भिन्नात्मक फलन हो तथा $x \in \mathbb{I}$

(D) $k(x) = \begin{cases} x^{\frac{1}{\ln x}} & \text{if } x \neq 1 \\ e & \text{if } x = 1 \end{cases}$, $x = 1$ पर

Column-II

(P) continuous

(Q) derivable

(R) discontinuous

(S) non derivable

Column-II

(P) सतत्

(Q) अवकलनीय

(R) असतत्

(S) अवकलनीय नहीं है।

Ans. (A) P, S; (B) P, Q; (C) R, S; (D) P, Q

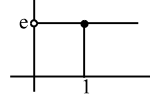
Sol. (A) $f'(0) = \lim_{h \rightarrow 0} \frac{\cosh - 0}{h}$ does not exist. Obviously $f(0) = f(0^-) = f(0^+) = 1$

Hence continuous and not derivable

(B) $g(x) = 0$ for all x , hence continuous and derivable

(C) as $0 \leq \{f(x)\} < 1$, hence $h(x) = \sqrt{\{x\}^2} = \{x\}$ which is discontinuous hence non derivable all $x \in I$

(D) $\lim_{x \rightarrow 1} x^{\frac{1}{\ln x}} = \lim_{x \rightarrow 1} x^{\log_x e} = e = f(1)$



61. Let $f(x) = \begin{cases} [x], & -2 \leq x < 0 \\ |x|, & 0 \leq x \leq 2 \end{cases}$ (where $[.]$ denotes the greatest integer function) $g(x) = \sec x$, $x \in \mathbb{R} - (2n+1)\frac{\pi}{2}$,

$n \in I$. Match the following statements in column I with their values in column II in the interval $\left(-\frac{3\pi}{2}, \frac{3\pi}{2}\right)$.

Column - I

- (A) Abscissa of points where limit of $\text{fog}(x)$ exist is/are
 (B) Abscissa points in domain of $\text{gof}(x)$, where limit of $\text{gof}(x)$ does not exist is/are
 (C) Abscissa of point of discontinuity of $\text{fog}(x)$ is/are
 (D) Abscissa of points of differentiability of $\text{fog}(x)$ is/are

Column-II

- (P) -1
 (Q) π
 (R) $\frac{5\pi}{6}$
 (S) $-\pi$
 (T) 0

माना $f(x) = \begin{cases} [x], & -2 \leq x < 0 \\ |x|, & 0 \leq x \leq 2 \end{cases}$ है (जहां $[.]$ महत्तम पूर्णांक फलन को दर्शाता है) $g(x) = \sec x$, $x \in \mathbb{R} - (2n+1)\frac{\pi}{2}$,
 $n \in I$

अन्तराल $\left(-\frac{3\pi}{2}, \frac{3\pi}{2}\right)$ में स्तम्भ I के मानों को स्तम्भ II से सूमेलित कीजिये -

स्तम्भ-I

- (A) x निर्देशांक जहां $\text{fog}(x)$ की सीमा विद्यमान है, होंगे
 (B) $\text{gof}(x)$ के प्रान्त में x निर्देशांक, जहां $\text{gof}(x)$ की सीमा विद्यमान नहीं है
 (C) x निर्देशांक जिस पर $\text{fog}(x)$ असतत है
 (D) x निर्देशांक जहां $\text{fog}(x)$ अवकलनीय है

स्तम्भ-II

- (P) -1
 (Q) π
 (R) $\frac{5\pi}{6}$
 (S) $-\pi$
 (T) 0

Ans. (A) PQRST, (B) PT, (C) QS, (D) PRT

Sol.

$$f(x) = \begin{cases} -2 & -2 \leq x < -1 \\ -1 & -1 \leq x < 0 \\ x & 0 \leq x \leq 2 \end{cases}$$

$$g(x) = \sec x; \quad x \in \mathbb{R} - \left\{ (2n+1)\frac{\pi}{2}; n \in \mathbb{I} \right\}$$

$$\therefore g(x) \text{ is continuous \& differentiable } \forall x \in \mathbb{R} - \left\{ (2n+1)\frac{\pi}{2} \right\} \text{ \& } f(x) \text{ is continuous } \forall x \in \mathbb{R} - \{-1, 0\}$$

$$\Rightarrow fog(x) \text{ is continuous } \forall x \in \mathbb{R} - \left\{ -1, 0, (2n+1)\frac{\pi}{2} \right\}$$

$$gof(x) \text{ is continuous } \forall x \in \mathbb{R} - \left\{ -1, 0, (2n+1)\frac{\pi}{2} \right\}$$

at $x = -1$

$$\lim_{x \rightarrow -1} f(\sec x) = f(\sec 1) = \sec 1 = fog(-1)$$

at $x = 0$

$$\lim_{x \rightarrow 0} f(\sec x) = f(1) = 1 = fog(0)$$

$\therefore fog(x)$ is continuous for $x = 0, -1$.

$$(B) \text{ Domain is } gof(x) = [-2, 2] - \left\{ \frac{\pi}{2}, -\frac{\pi}{2} \right\}$$

$\therefore g(f(x))$ is discontinuous at $x = -1, 0$.

at $x = \pi$

$$\lim_{h \rightarrow 0} fog(\pi + h) = \lim_{h \rightarrow 0} f(\sec h) = -1$$

$$\lim_{h \rightarrow 0} fog(\pi - h) = \lim_{h \rightarrow 0} f(-\sec h) = -2$$

at $x = -\pi$

$$LHL = -2, RHL = -1$$

$\therefore f(x)$ is discontinuous at $x = -\pi, \pi$

Exercise J-M

1. Let a function $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} \sin x - e^x & \text{if } x \leq 0 \\ a + [-x] & \text{if } 0 < x < 1 \\ 2x - b & \text{if } x \geq 1 \end{cases}$$

where $[x]$ is the greatest integer less than or equal to x . If f is continuous on $\mathbb{R}$, then $(a + b)$ is equal to :

[JEE(Main) July 2021]

माना एक फलन $f: \mathbb{R} \rightarrow \mathbb{R}$

$$f(x) = \begin{cases} \sin x - e^x & \text{यदि } x \leq 0 \\ a + [-x] & \text{यदि } 0 < x < 1 \\ 2x - b & \text{यदि } x \geq 1 \end{cases}$$

द्वारा परिभाषित है, जहाँ $[x]$ महत्तम पूर्णांक $\leq x$ है। यदि फलन f , $\mathbb{R}$ पर संतत है, तो $(a + b)$ बराबर है :

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(A) 2

(B) 5

(C) 4

(D) 3

Ans. D

Sol. (i) Continuous at $x = 0$

$$\text{LHL} = -1$$

$$f(0) = -1$$

$$\text{RHL} = a - 1$$

$$a - 1 = -1$$

$$\Rightarrow a = 0$$

(ii) Continuous at $x = 1$

$$\text{LHL} = a - 1$$

$$f(1) = 2 - b$$

$$\text{RHL} = 2 - b$$

$$a - 1 = 2 - b$$

$$\Rightarrow b = 3$$

2. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as
$$f(x) = \begin{cases} \frac{x^3}{(1 - \cos 2x)^2} & \log_e \left(\frac{1 + 2xe^{-2x}}{(1 - xe^{-x})^2} \right), x \neq 0. \\ \alpha & , x = 0 \end{cases}$$

If f is continuous at $x = 0$, then α is equal to :

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